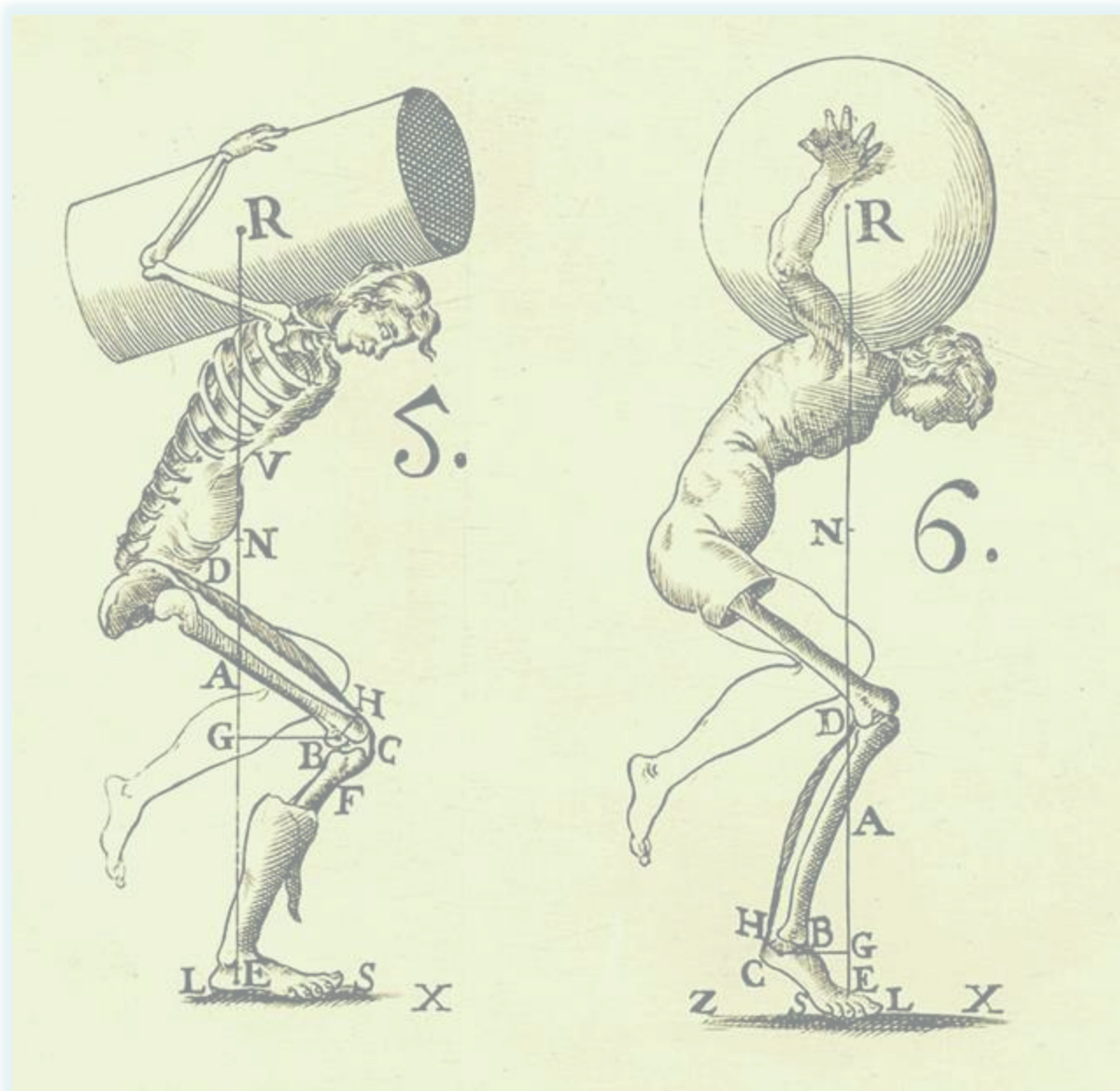






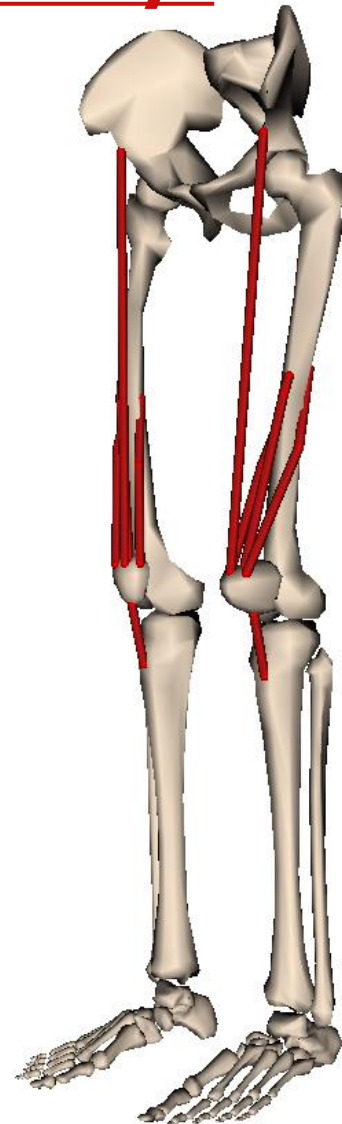
Chapter 6: Musculoskeletal Geometry

Plan for today

- Musculoskeletal geometry
 - Motivation
 - Definition of a moment (and moment arm)
 - 2D and 3D examples
 - How to measure moment arms
 - Example applications
- Joint moments
- Modeling of geometry

Why study musculoskeletal geometry?

- To answer fundamental questions:
 - Why are we shaped this way and how do variations in geometry affect performance?
 - What if we had huge moment arms?

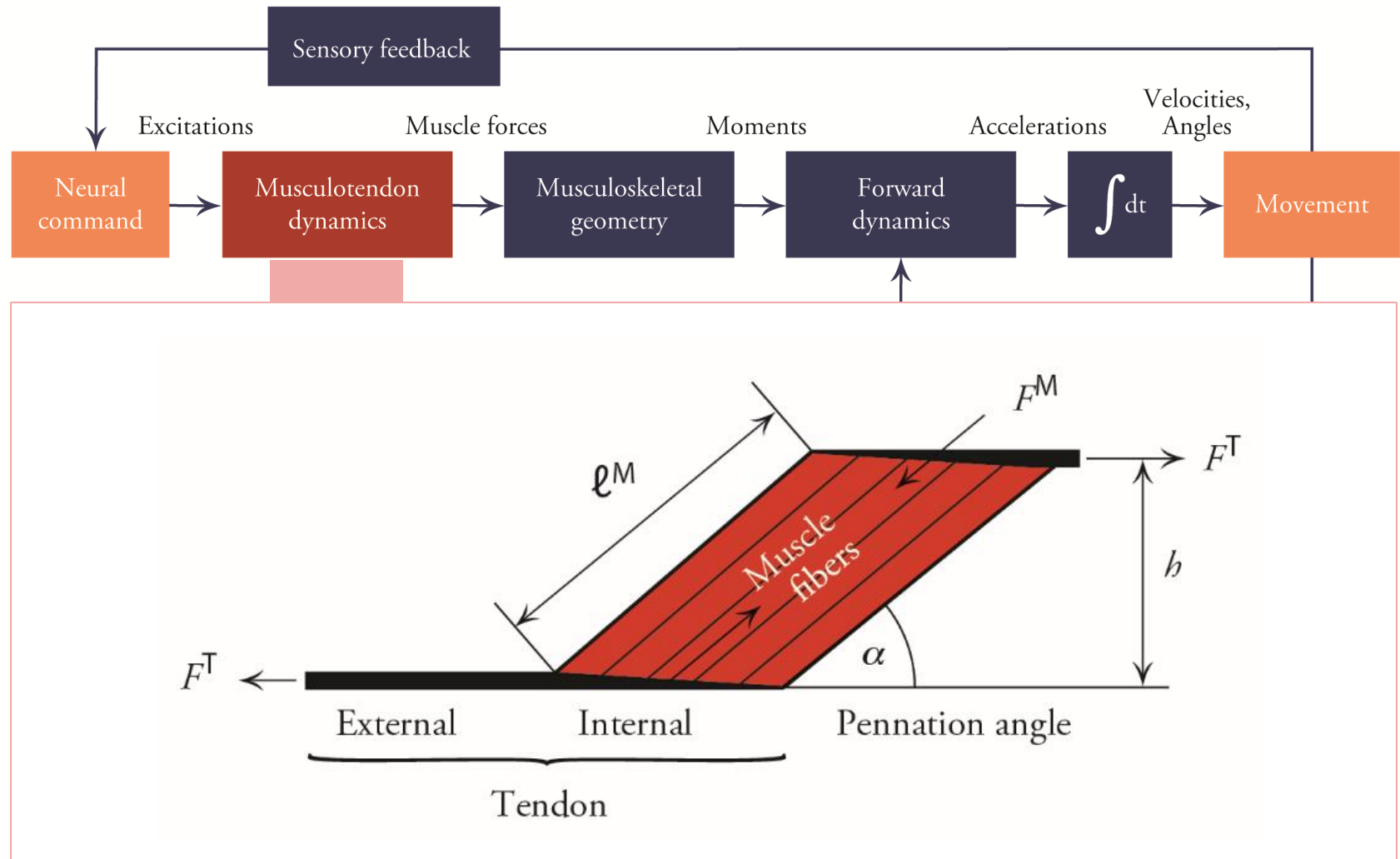


Why study musculoskeletal geometry?

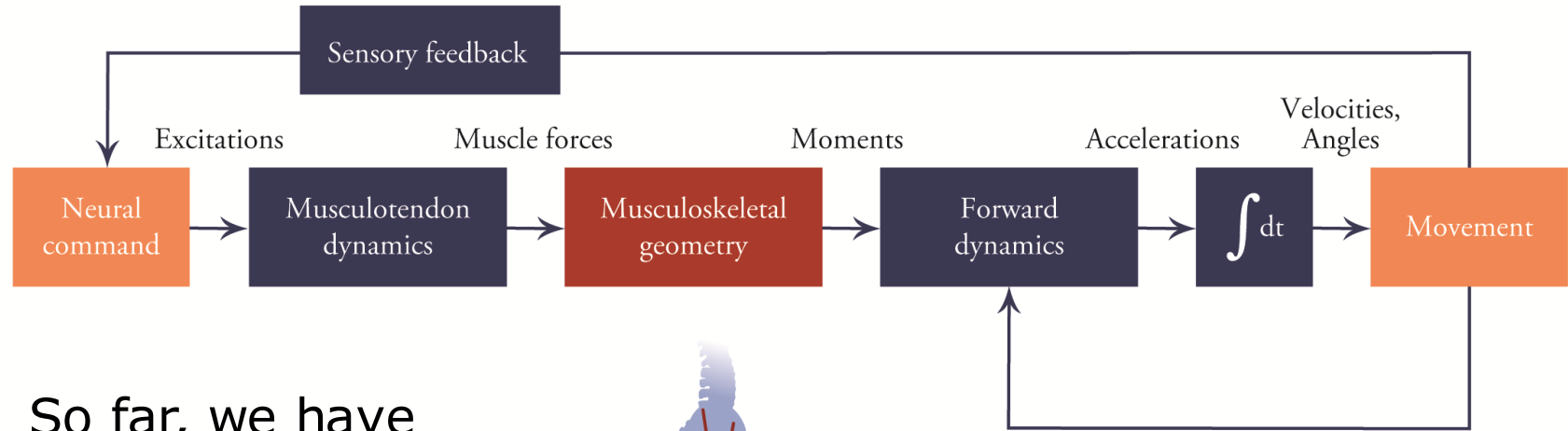
- To answer fundamental questions:
 - Why are we shaped this way and how do variations in geometry affect performance?
 - What if we had huge moment arms?
- To answer clinical questions:
 - How is muscle function altered by deformity?
 - How can we design surgical changes in geometry to restore function?



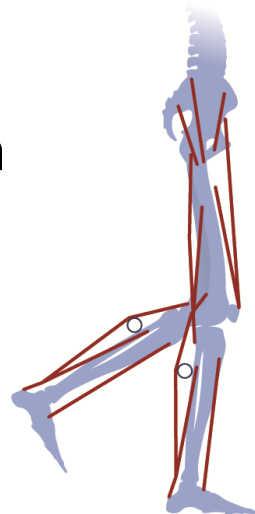
Musculotendon dynamics transform neural command to force



Musculoskeletal geometry transforms these tendon forces to joint moments



So far, we have analyzed muscles in isolation, but musculoskeletal geometry has a major influence on muscle function.



Moment arms are the basic measure of geometry.

Definition of a moment

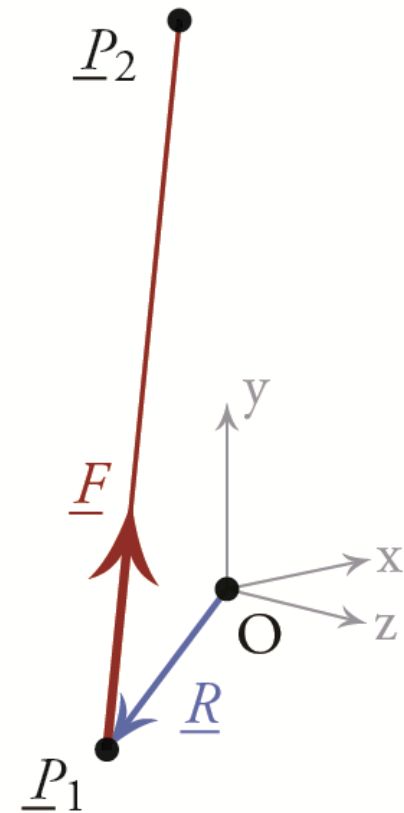
“Moment ($\vec{M}_O$) of a force ($\vec{F}$) about point O .”

The force, $\vec{F}$, acts at a point, P_1 , along a line of action (e.g., from P_1 to P_2).

Let $\vec{R}$ be a vector from O to any point on the line of action of $\vec{F}$.

The moment, $\vec{M}_O$, of $\vec{F}$ about O is defined by the cross-product

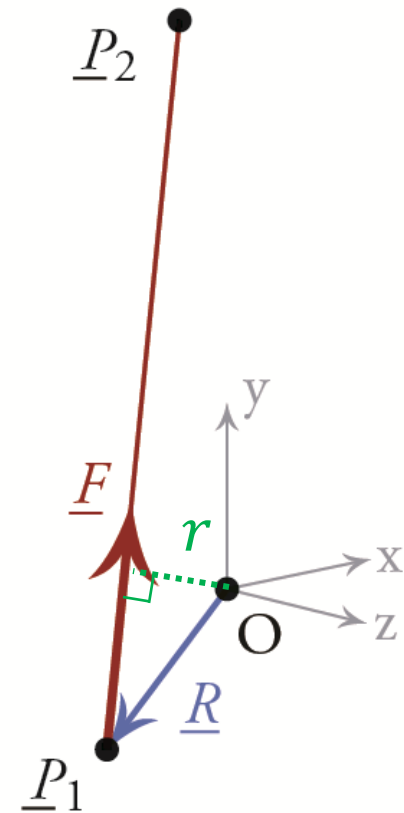
$$\vec{M}_O \equiv \vec{R} \times \vec{F}$$



Definition of a moment arm

The moment arm, r , of $\vec{F}$ about O for the $\hat{z}$ -direction is

$$r = \frac{M_{O,z}}{|\vec{F}|} = \frac{\vec{R} \times \vec{F} \cdot \hat{z}}{|\vec{F}|}$$

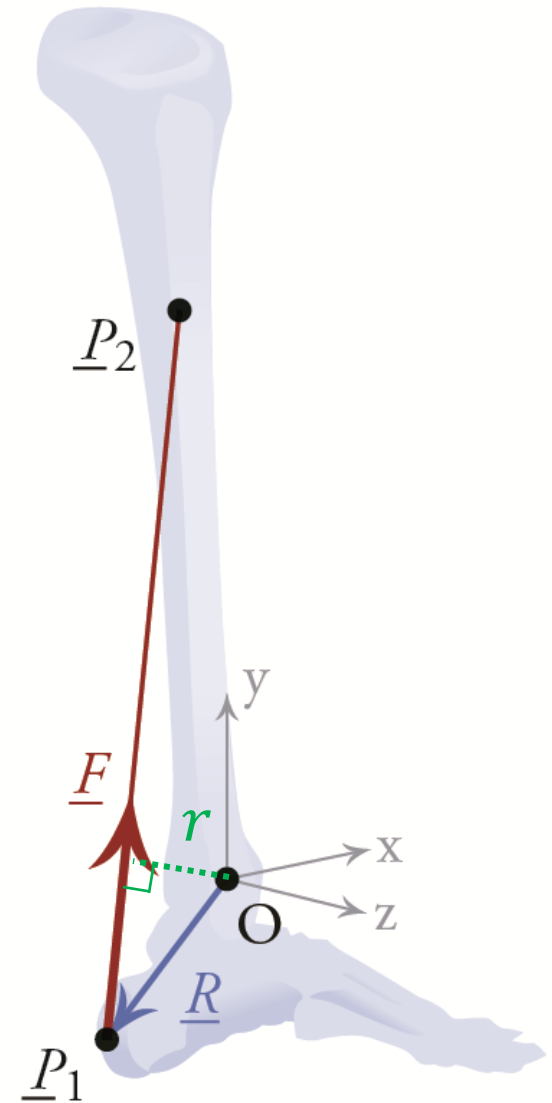


Definition of a moment arm

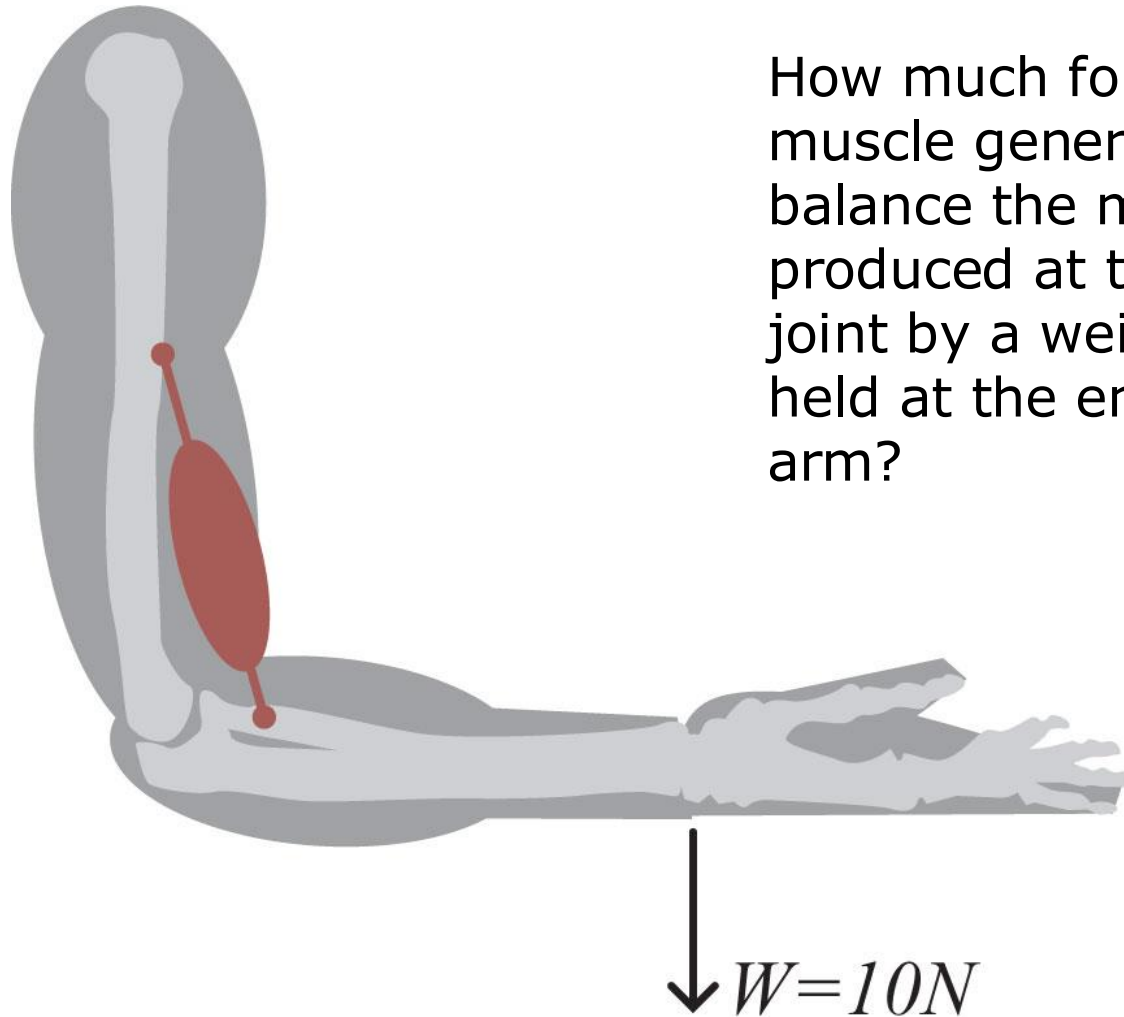
The moment arm, r , of $\vec{F}$ about O for the $\hat{z}$ -direction is

$$r = \frac{M_{O,z}}{|\vec{F}|} = \frac{\vec{R} \times \vec{F}}{|\vec{F}|} \cdot \hat{z}$$

Where do we get this geometry?

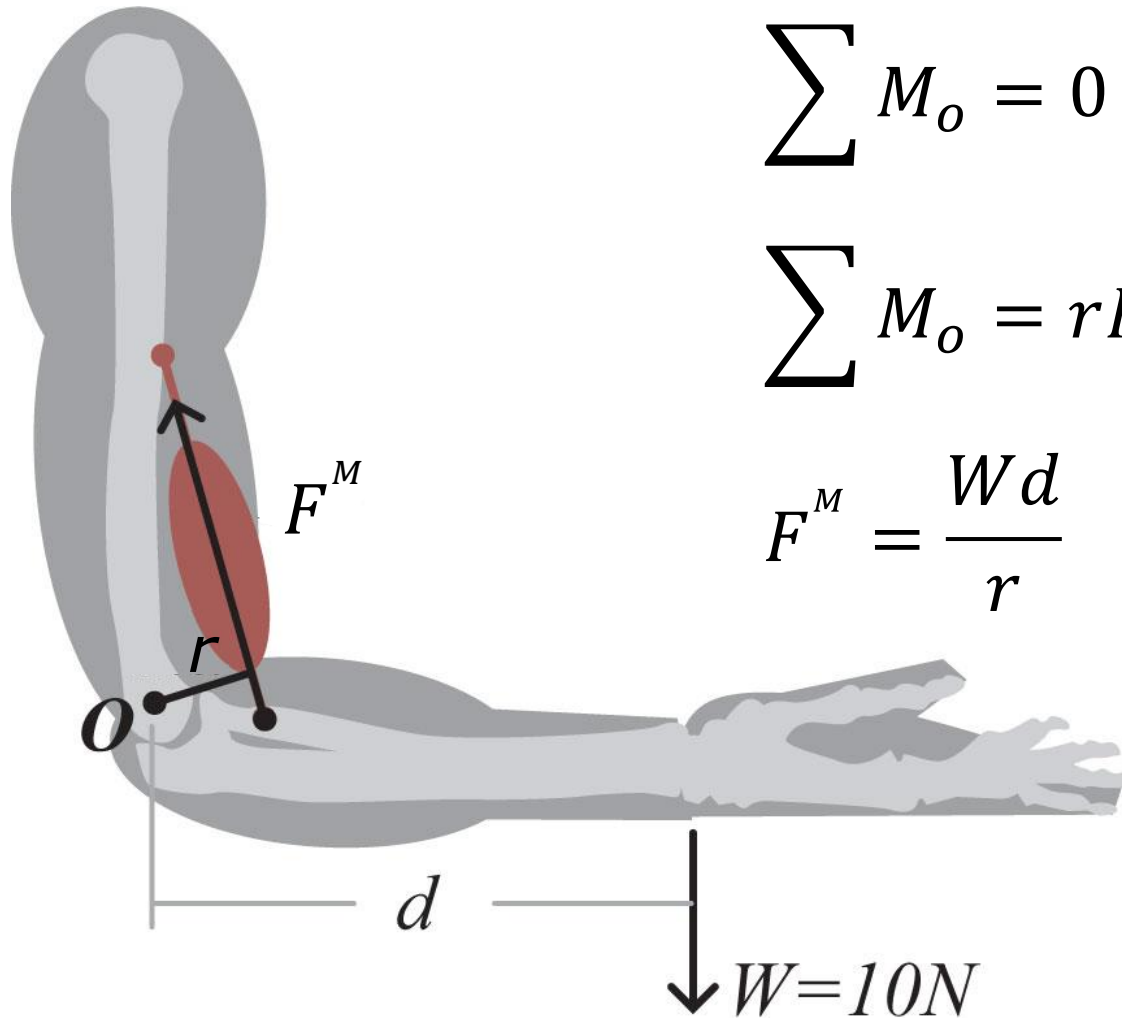


Planar isometric static analysis



How much force must a muscle generate to balance the moment produced at the elbow joint by a weight being held at the end of the arm?

Planar isometric static analysis



$$\sum M_o = 0$$

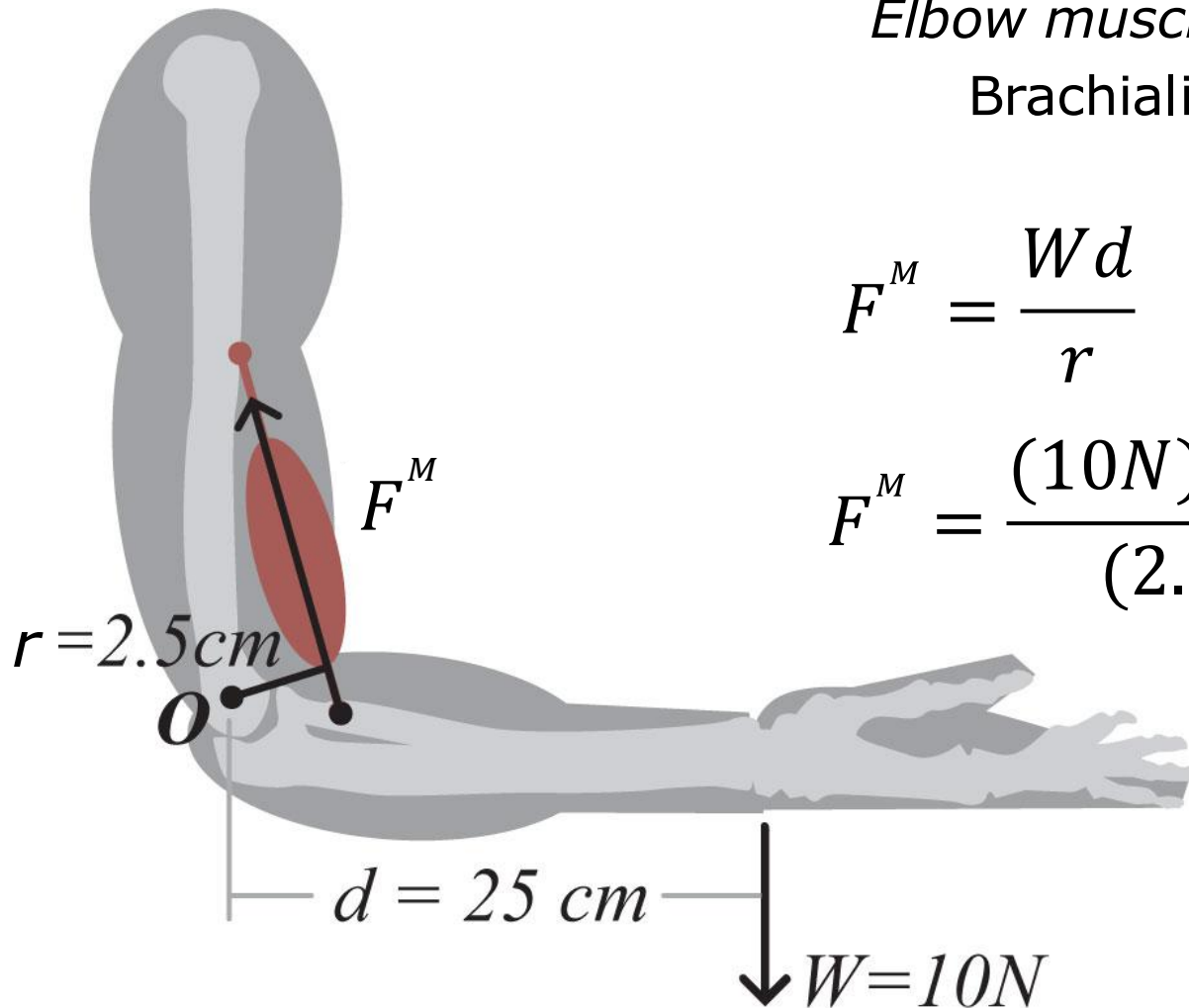
$$\sum M_o = rF^M - Wd = 0$$

$$F^M = \frac{Wd}{r}$$

What assumptions did we make?

Planar isometric static analysis

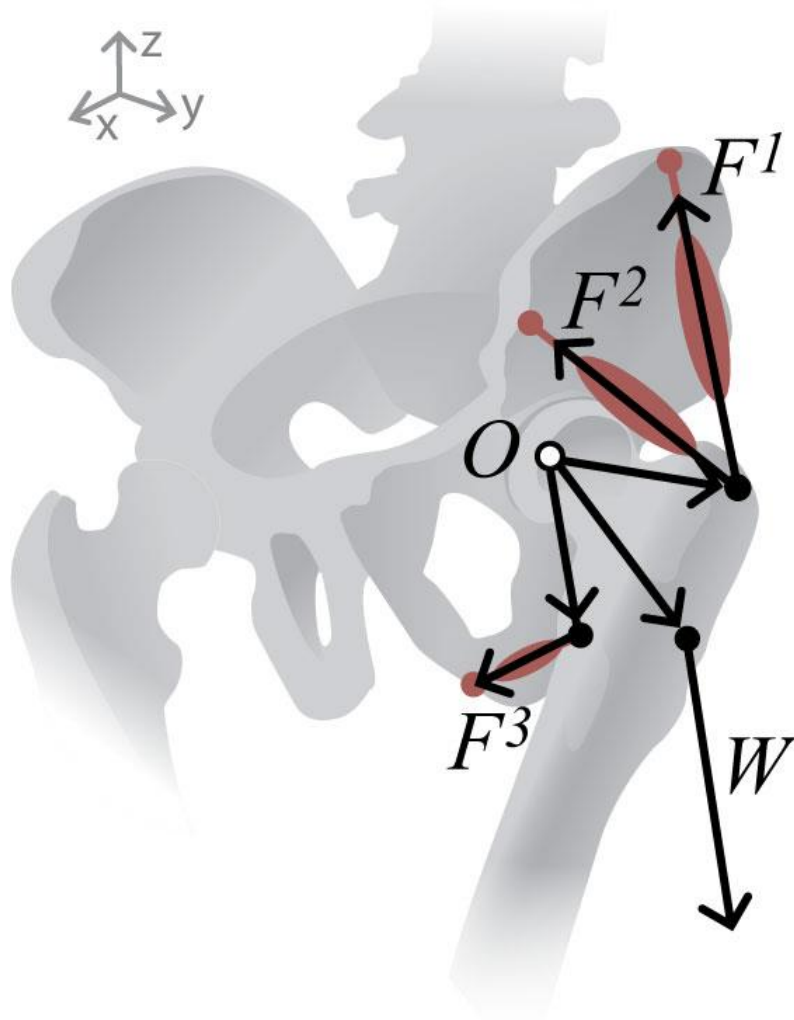
Elbow muscle moment arm:
Brachialis ~ 2.5 cm



$$F^M = \frac{Wd}{r}$$

$$F^M = \frac{(10 \text{ N})(25 \text{ cm})}{(2.5 \text{ cm})} = 100 \text{ N}$$

3D isometric static analysis



$$\sum \vec{M}_o = \vec{0}$$

$$(\vec{R}_w \times \vec{W}) + \sum_{i=1}^3 (\vec{R}_i \times \vec{F}^i) = \vec{0}$$

$$(\vec{R}_w \times \vec{W}) \cdot \hat{x} + \sum_{i=1}^3 (\vec{R}_i \times \vec{F}^i) \cdot \hat{x} = 0$$

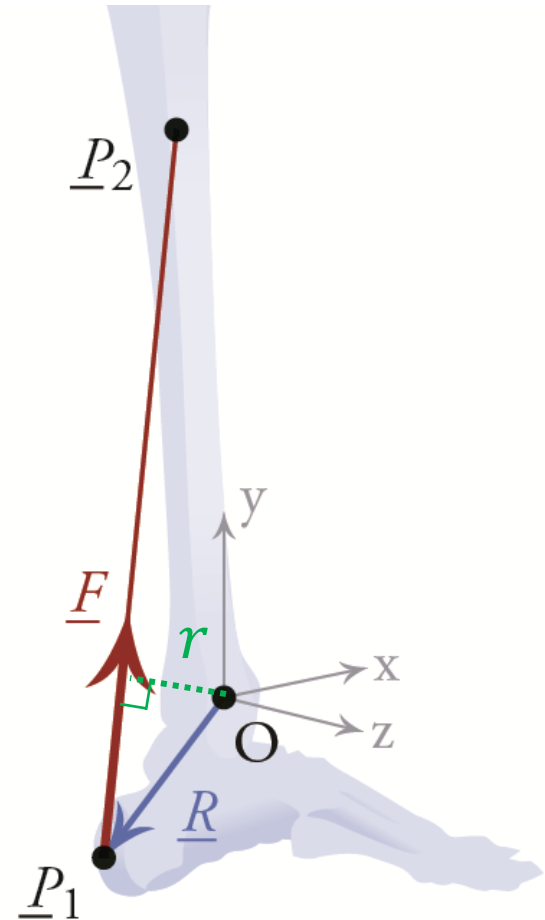
$$(\vec{R}_w \times \vec{W}) \cdot \hat{y} + \sum_{i=1}^3 (\vec{R}_i \times \vec{F}^i) \cdot \hat{y} = 0$$

$$(\vec{R}_w \times \vec{W}) \cdot \hat{z} + \sum_{i=1}^3 (\vec{R}_i \times \vec{F}^i) \cdot \hat{z} = 0$$

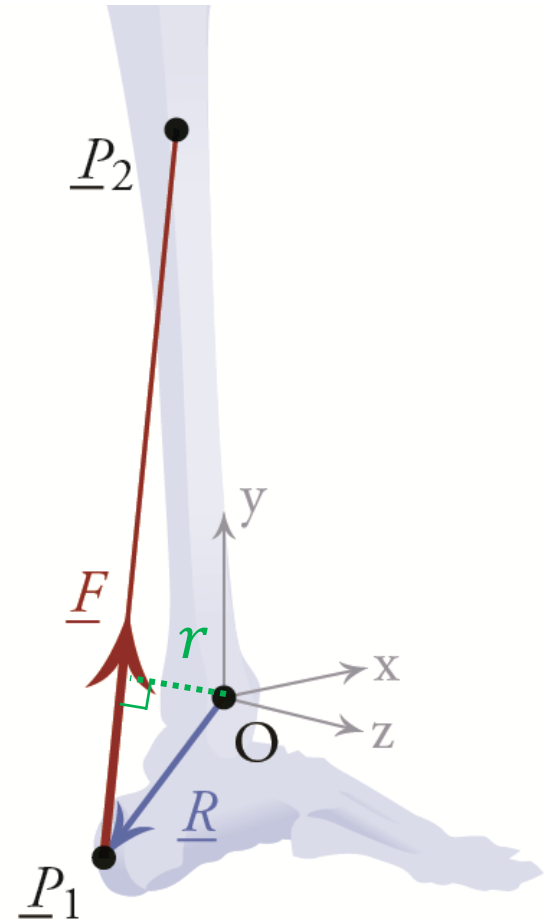
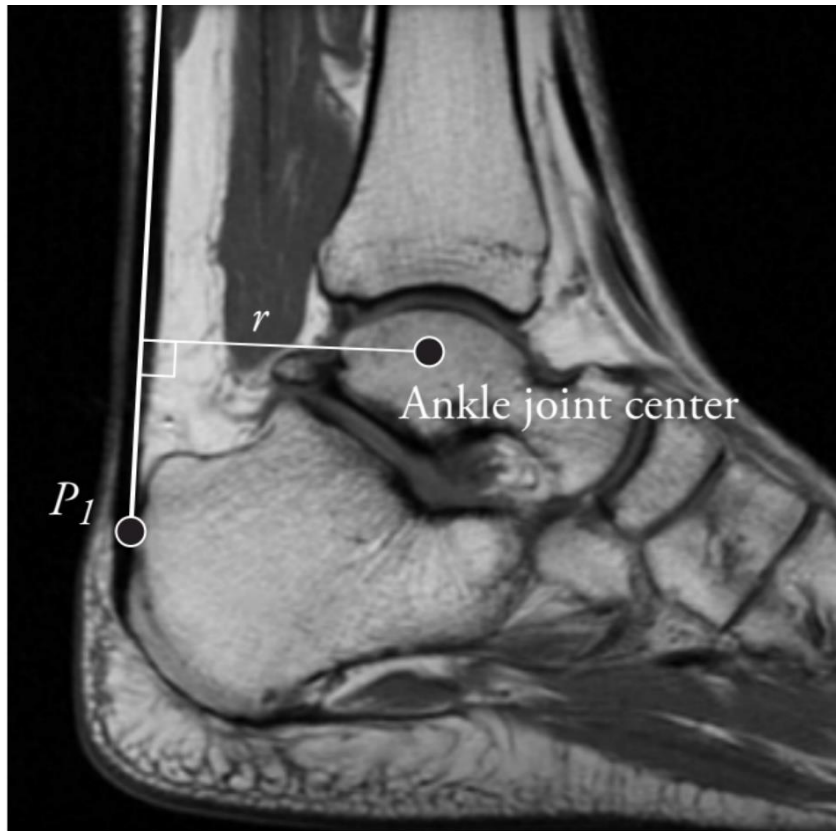
How do we measure moment arms?

The moment arm, r , of $\vec{F}$ about O for the $\hat{z}$ -direction is

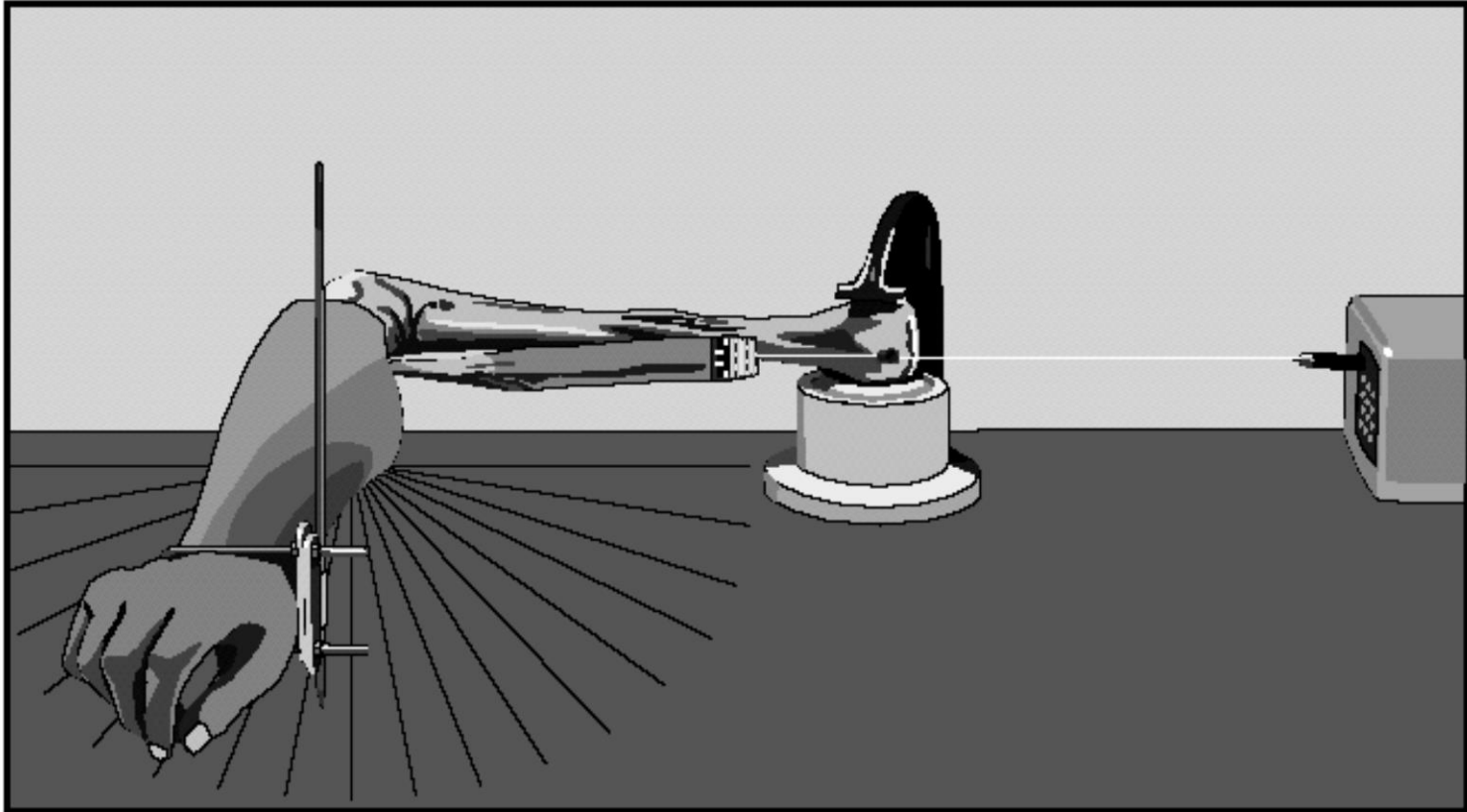
$$r = \frac{M_{O,z}}{|\vec{F}|} = \frac{\vec{R} \times \vec{F} \cdot \hat{z}}{|\vec{F}|}$$



Moment arms via imaging



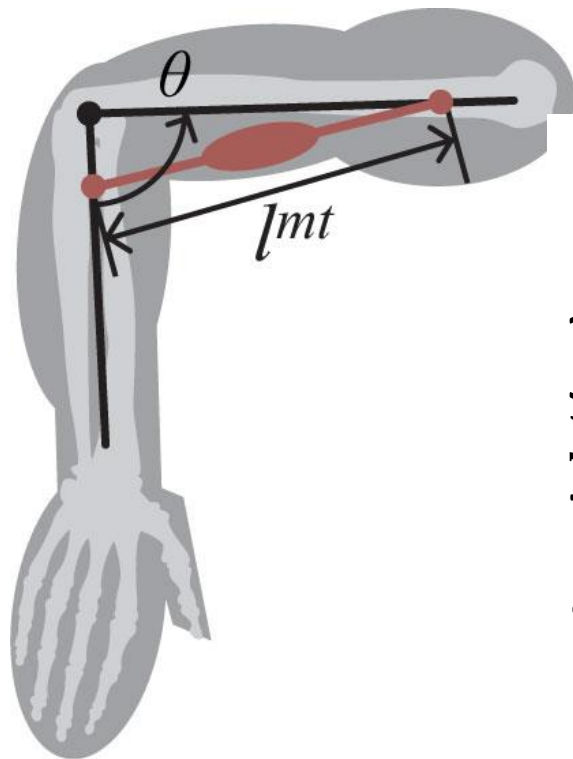
Tendon excursion method relates change in muscle-tendon length to change in joint angle.



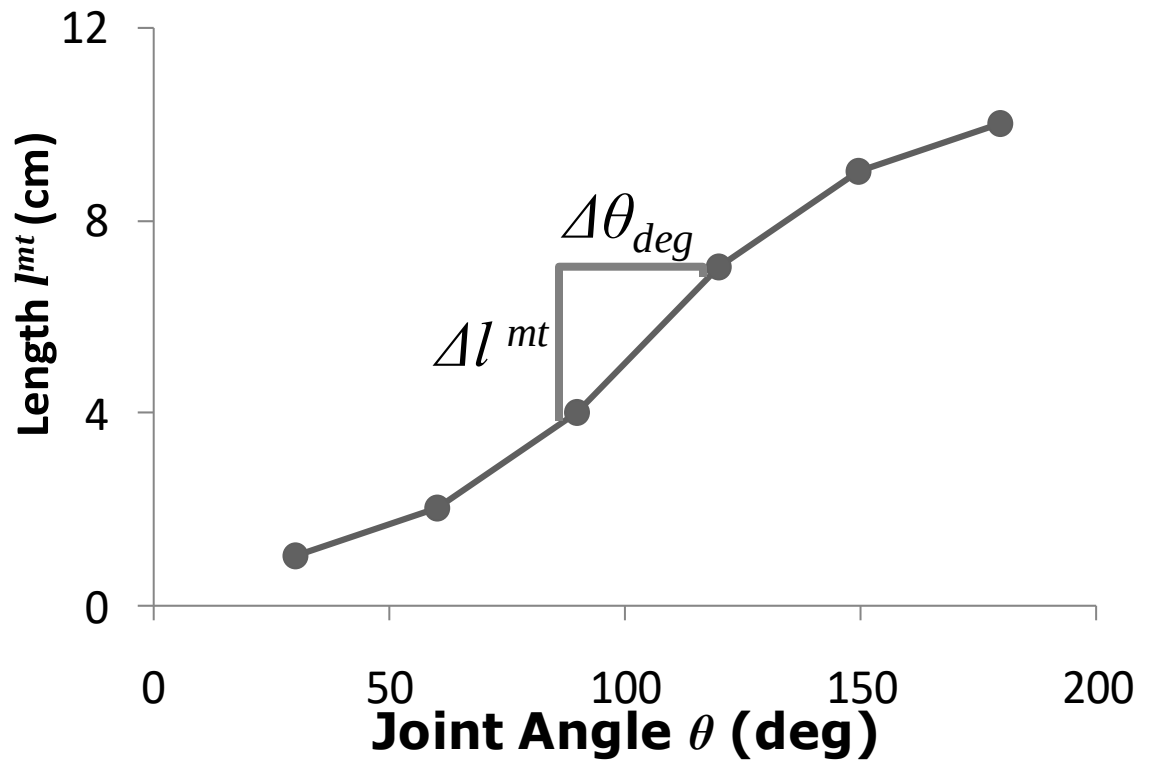
Murray, Delp and Buchanan, 1995

You'll show in homework: $r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$

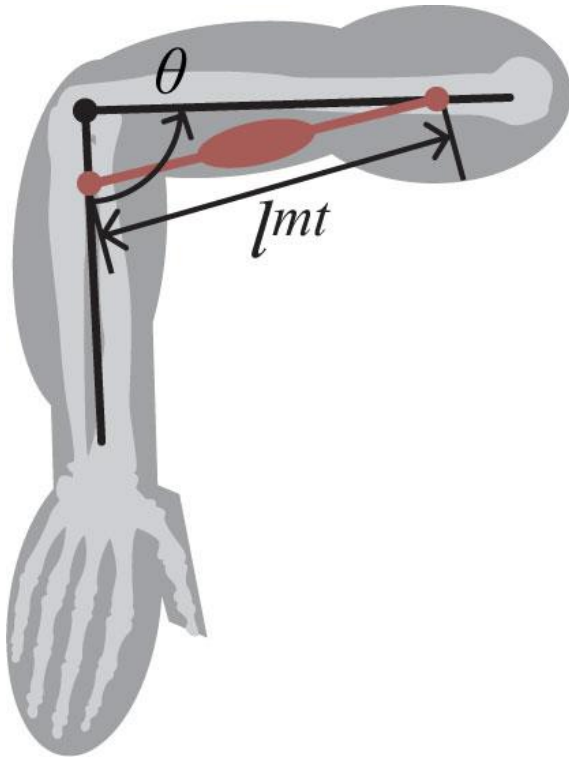
Muscle-Tendon Length vs. Joint Angle



$$r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$$



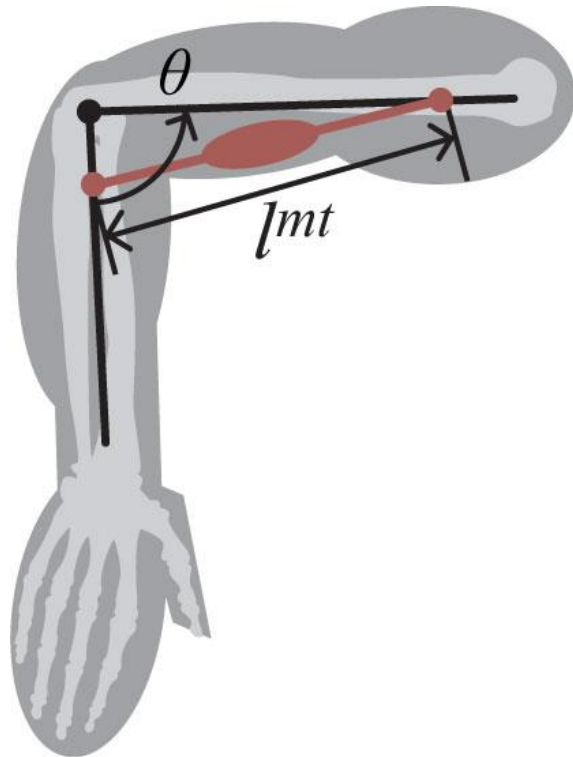
Sample Data: Joint Angle and Length



θ_{deg}	$\Delta\theta_{deg}$	$\Delta\theta_{rad}$	l^{mt}_{cm}	Δl^{mt}_{cm}	r_{cm}
30			1		
60			2		
90			4		
120			7		
150			9		
180			10		

$$r = \frac{\Delta l^{MT}}{\Delta\theta_{rad}}$$

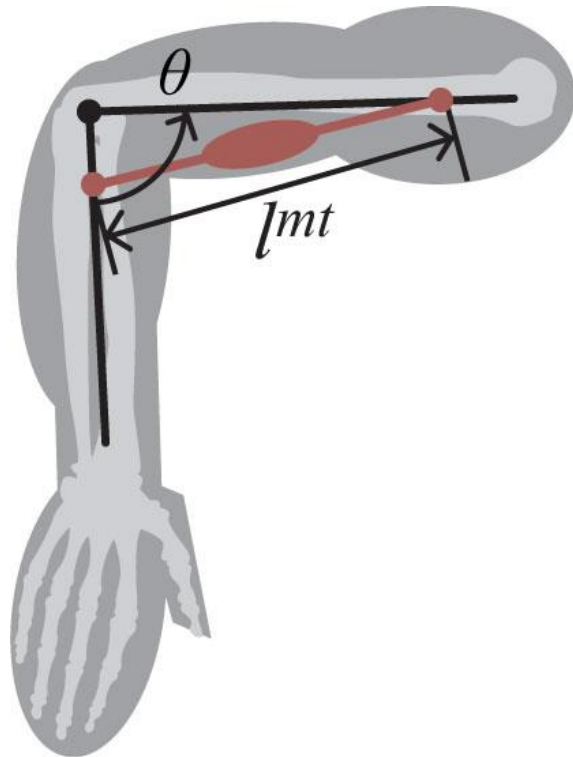
Sample Data: Joint Angle and Length



$$r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$$

θ_{deg}	$\Delta \theta_{deg}$	$\Delta \theta_{rad}$	l^{mt}_{cm}	Δl^{mt}_{cm}	r_{cm}
30	) → 30		1	) → 1	
60			2		
90			4		
120			7		
150			9		
180			10		

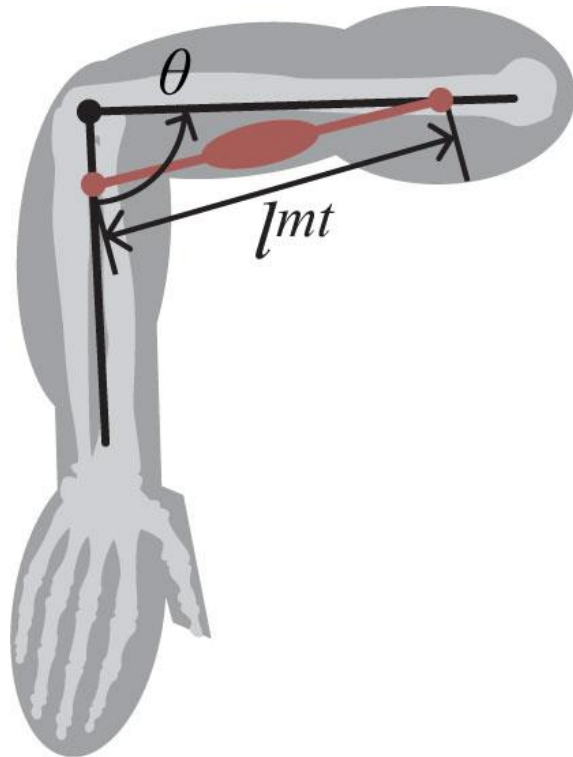
Sample Data: Joint Angle and Length



$$r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$$

θ_{deg}	$\Delta \theta_{deg}$	$\Delta \theta_{rad}$	$l_{mt\ cm}$	$\Delta l_{mt\ cm}$	r_{cm}
30	) → 30	0.5	1	) → 1	
60			2		
90	) → 30	0.5	4	) → 2	
120			7		
150	) → 30	0.5	9	) → 2	
180			10		

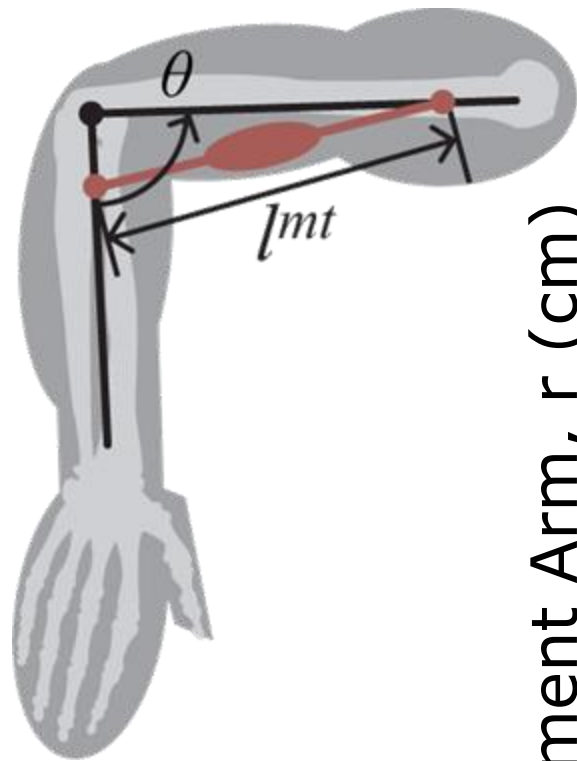
Sample Data: Joint Angle and Length



$$r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$$

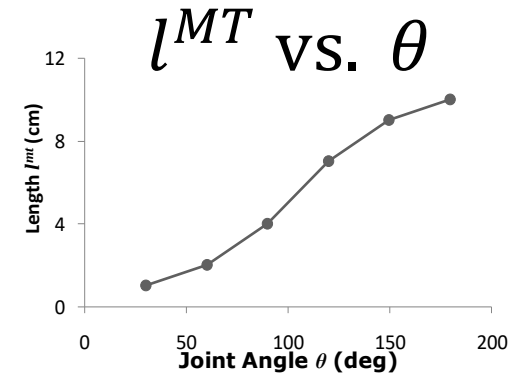
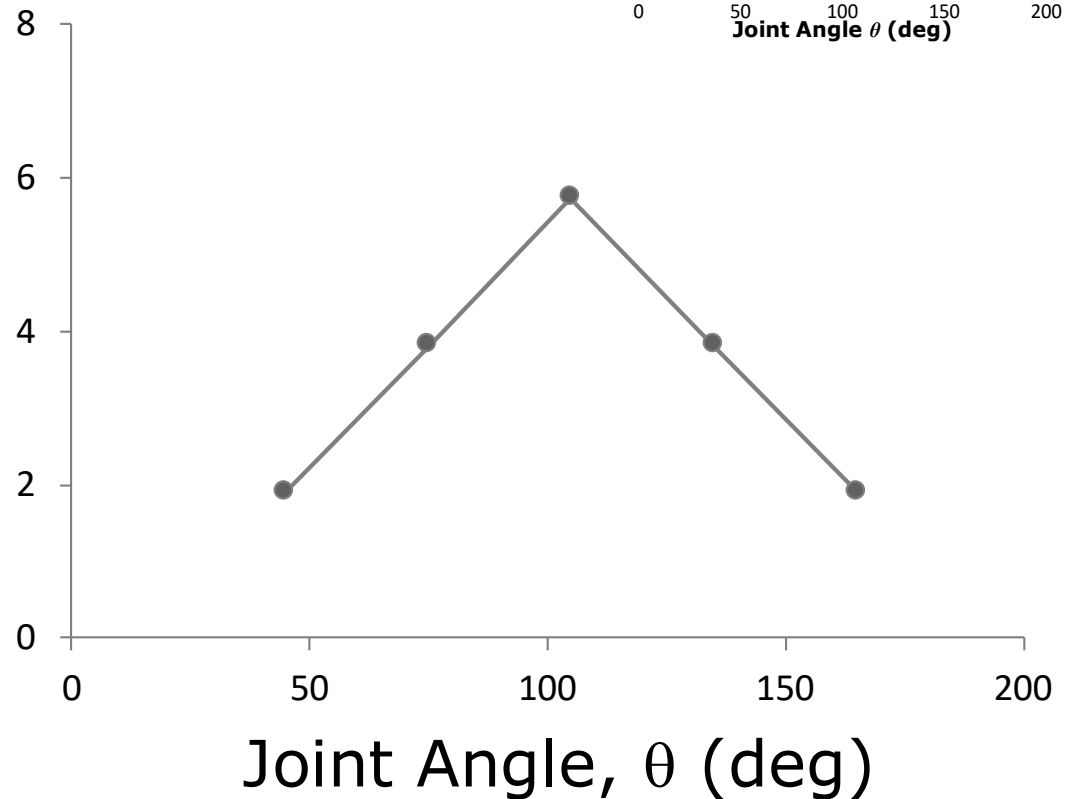
θ_{deg}	$\Delta \theta_{deg}$	$\Delta \theta_{rad}$	$l_{mt\ cm}$	$\Delta l_{mt\ cm}$	r_{cm}
30	) → 30	0.5	1	) → 1	2
60			2		
90	) → 30	0.5	4	) → 2	4
120			7		
150	) → 30	0.5	9	) → 2	4
180			10		
	) → 30	0.5		) → 1	2

Moment Arm vs. Joint Angle



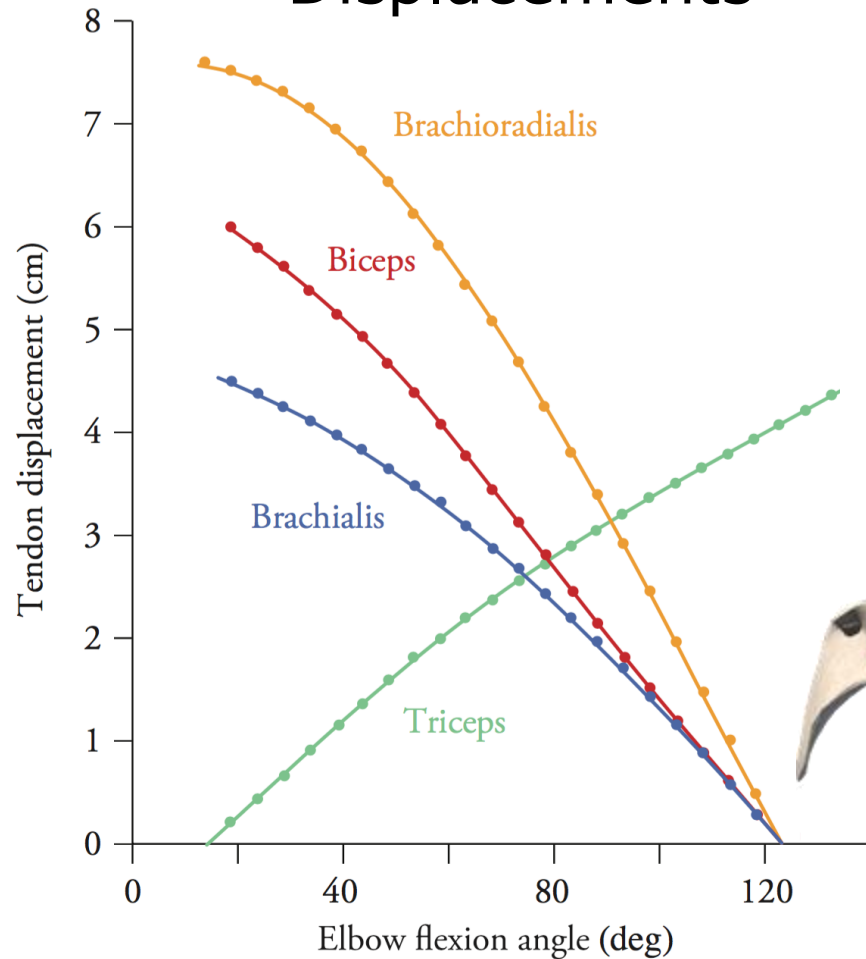
$$r = \frac{\Delta l^{MT}}{\Delta \theta_{rad}}$$

Moment Arm, r (cm)

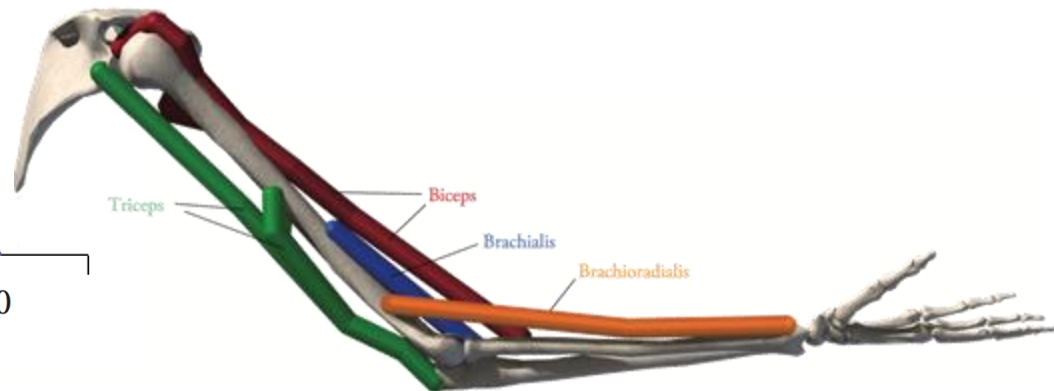


Experimental Results

Displacements

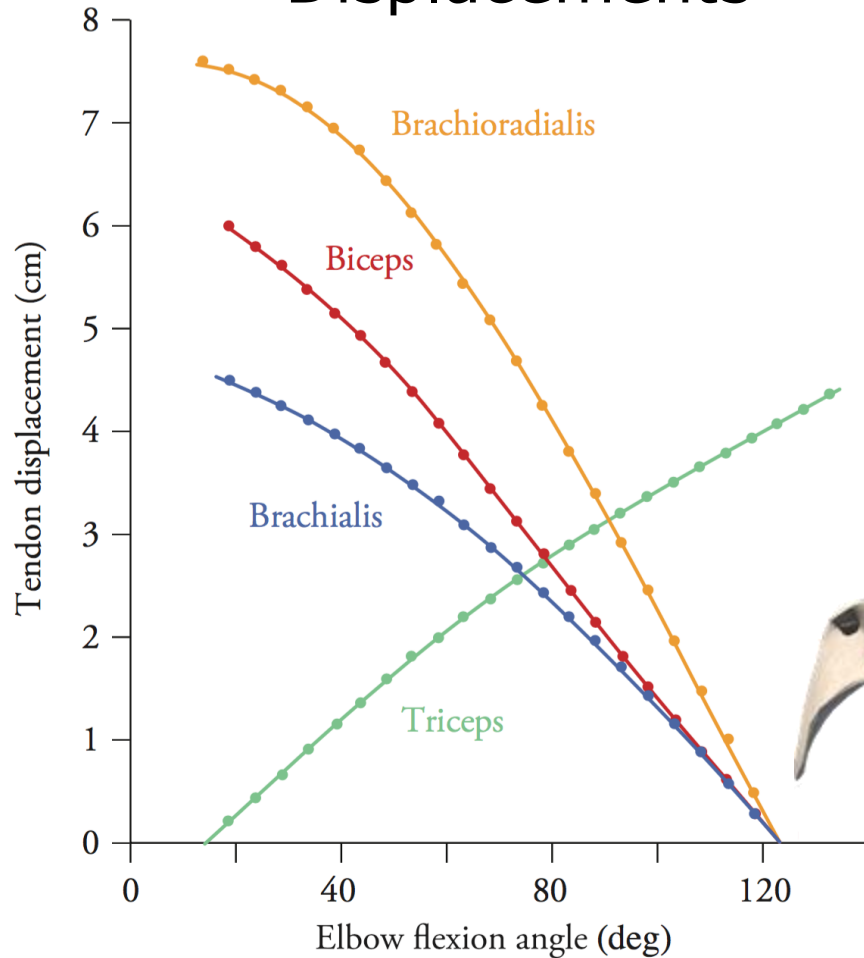


Wendy Murray

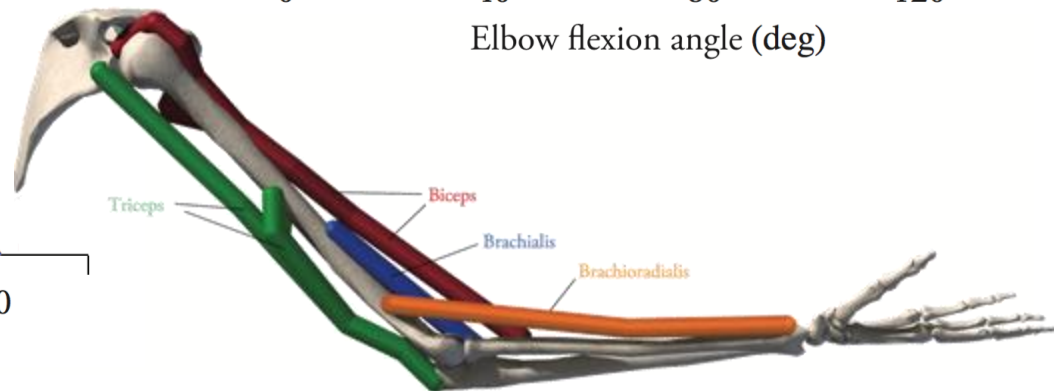
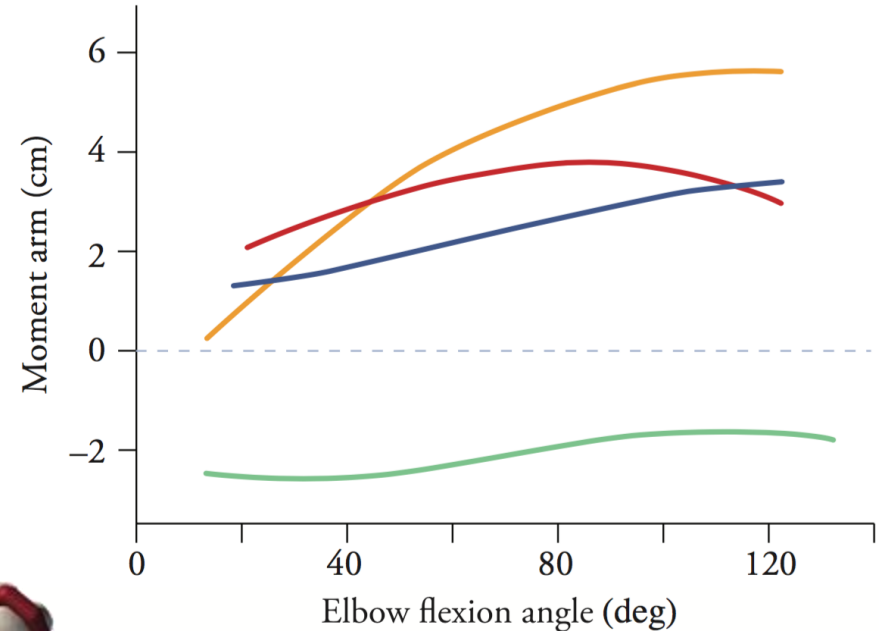


Experimental Results

Displacements



Moment Arms



Concept test

Muscles with large moment arms undergo a larger change in length with joint rotation.

True or False?

Draw this and convince yourself and your neighbor.

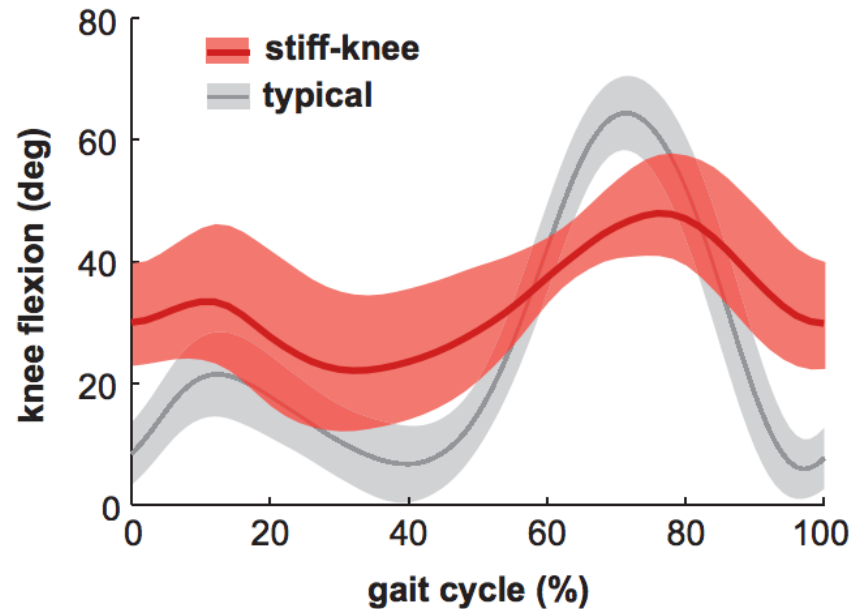
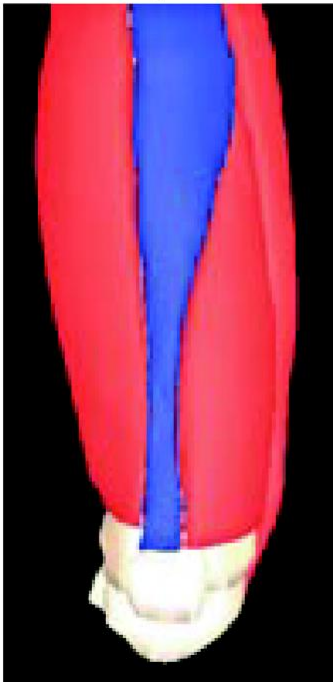
Plan for today

- Musculoskeletal geometry
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Application: Tendon Transfer

Example: Rectus Femoris Transfer to Improve Stiff Knee Gait

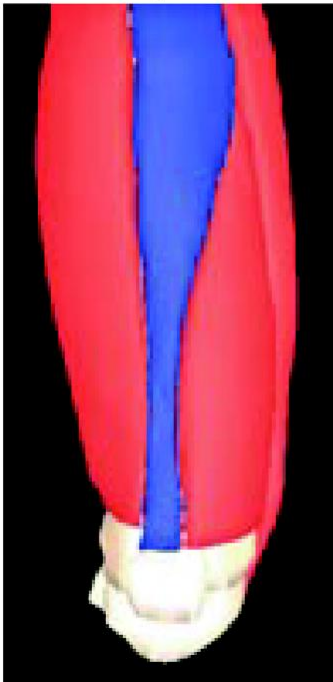
Non-Operated



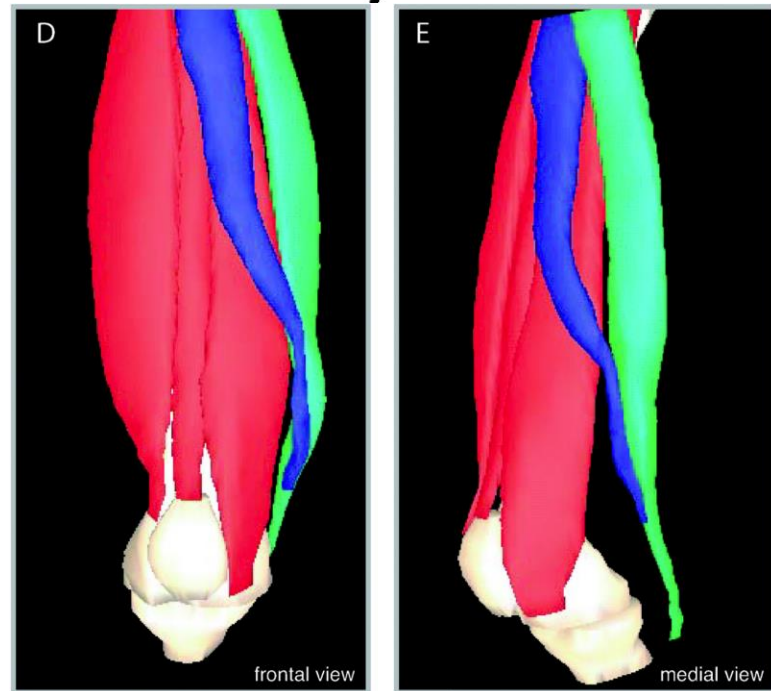
Application: Tendon Transfer

Example: Rectus Femoris Transfer to Improve Stiff Knee Gait

Non-Operated



Subject 1



Application: Tendon Transfer

Donor Muscle:

$$F_o^m = 1200 N$$

$$l_o^m = 6 cm$$

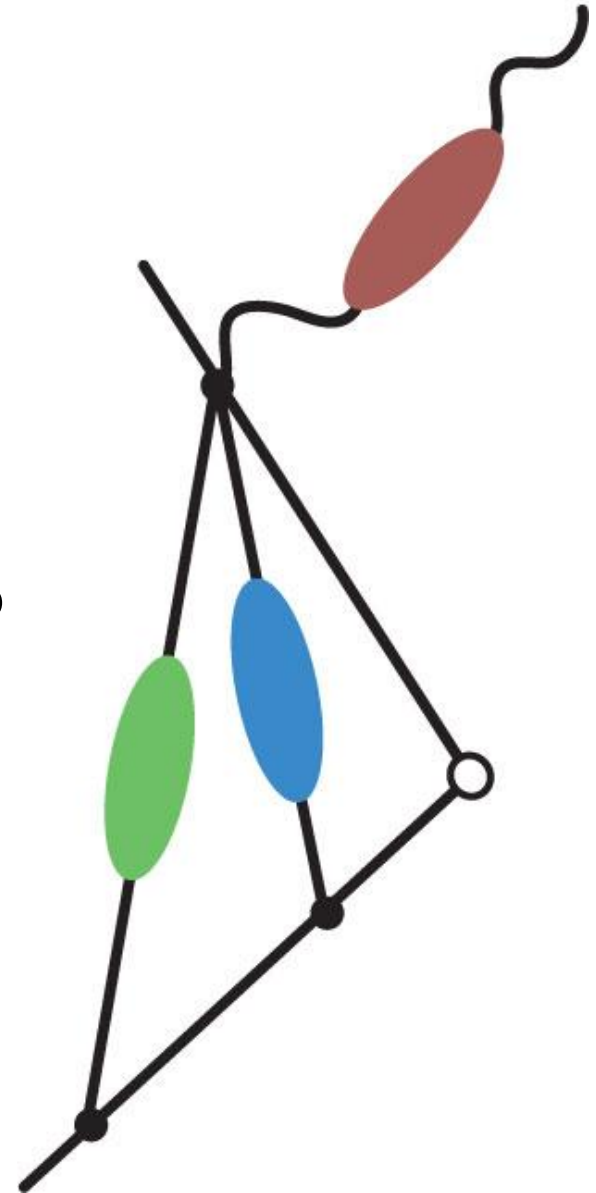
Surgeon must choose between two muscle “recipients”:

Muscle A: $r_A = 3 \text{ cm}$

Muscle B: $r_B = 6 \text{ cm}$

Requirements:

Muscle needs to produce a 10 Nm active moment over a 60° ROM to restore desired function

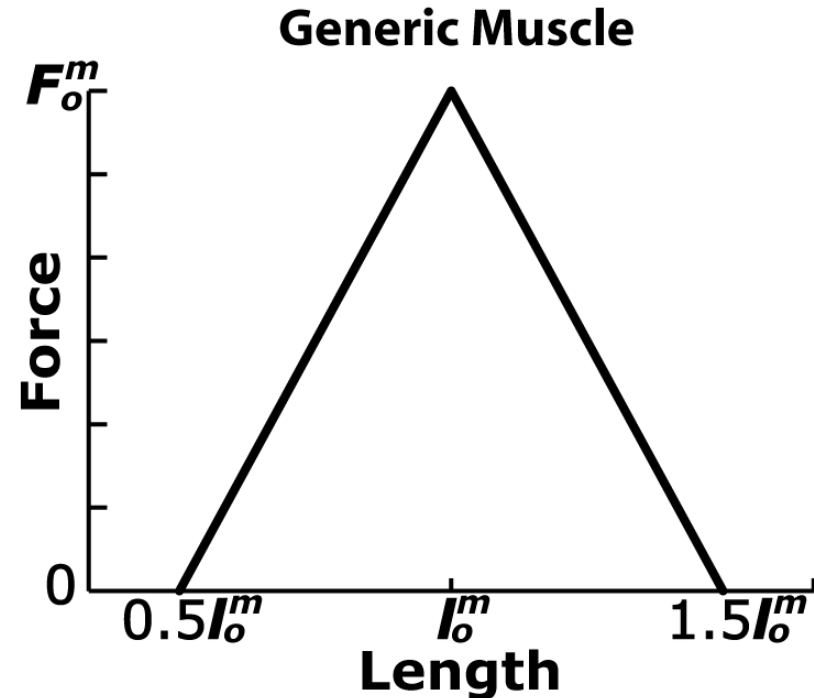


Application: Tendon Transfer

Requirements:

10 Nm over 60° ROM

Will attaching the donor to Muscle A do the job?



Application: Tendon Transfer

Requirements:

10 Nm over 60° ROM

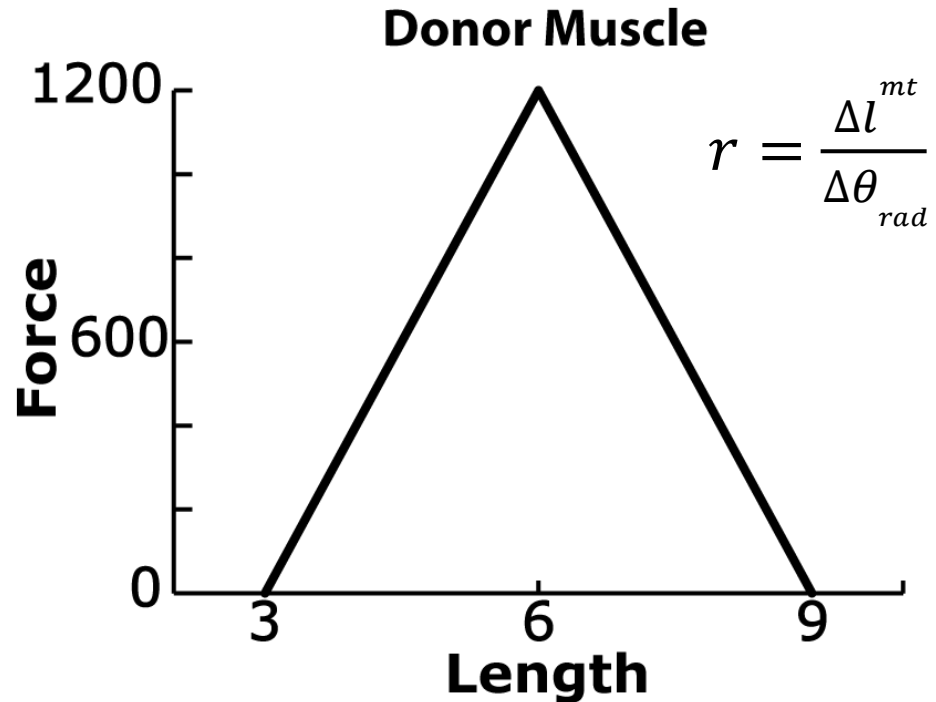
Will attaching the donor to Muscle A do the job?

Donor muscle

Properties :

$$F_o^m = 1200 \text{ N}$$

$$l_o^m = 6 \text{ cm}$$



Application: Tendon Transfer

Requirements:

10 Nm over 60° ROM

Will attaching the donor to Muscle A do the job?

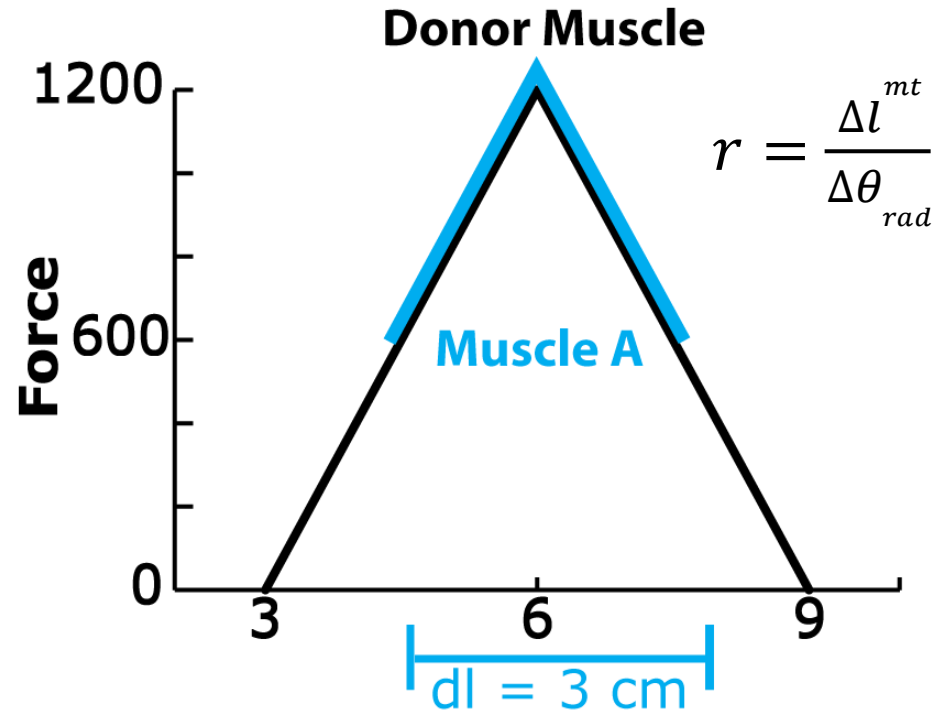
$$r_A = 3 \text{ cm}$$

Donor muscle

Properties :

$$F_o^m = 1200 \text{ N}$$

$$l_o^m = 6 \text{ cm}$$



$$M_{min,A} = r_a F_{min,A}$$

$$M_{min,A} = (0.03 \text{ m})(600 \text{ N}) = 18 \text{ Nm}$$

Application: Tendon Transfer

Requirements:

10 Nm over 60° ROM

Will attaching the donor to Muscle B do the job?

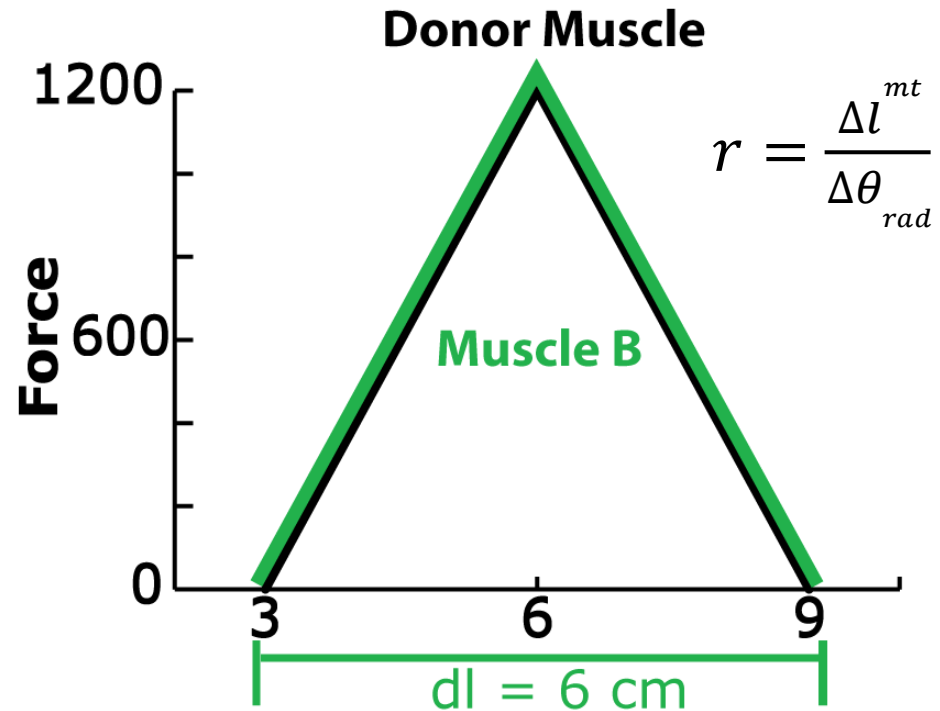
$r_b = 6 \text{ cm}$

Donor muscle

Properties :

$$F_o^m = 1200 \text{ N}$$

$$l_o^m = 6 \text{ cm}$$



$$M_{min,B} = r_b F_{min,B}$$

$$M_{min,B} = (0.06 \text{ m})(0 \text{ N}) = 0 \text{ Nm}$$

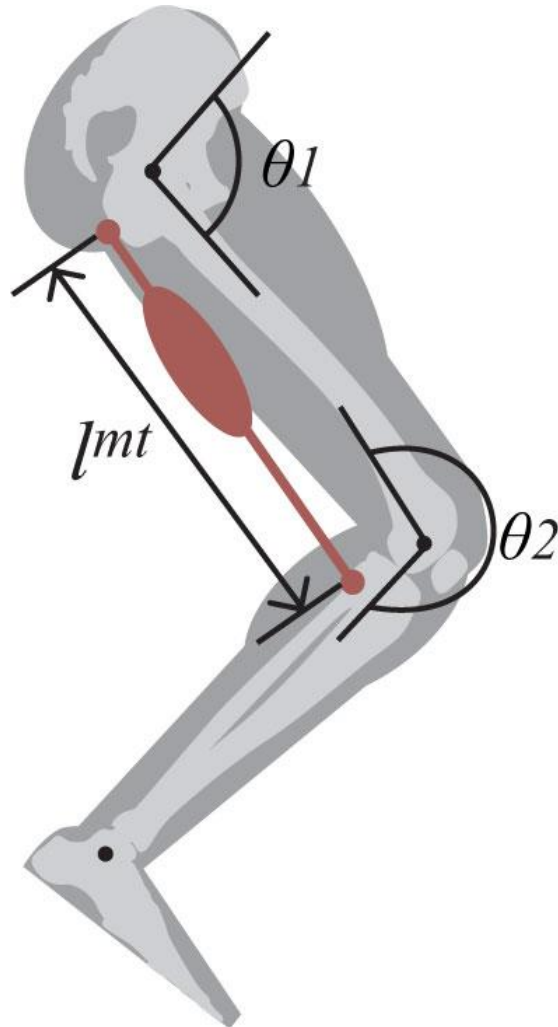
Recap – Tendon Transfer

Pros and cons of attaching to Muscle A vs Muscle B?

Extension- what if recipient muscle's moment arm was not constant?

We assumed the tendon was rigid (i.e. all changes in muscle-tendon unit length, we attributed to the muscle fiber.)

Application: Muscle-Tendon Velocity During Sprinting



$$\frac{dl^{mt}}{d\theta_i} = r_i$$

$$\cancel{\frac{dl^{mt}}{d\theta_i}} \cancel{\frac{d\theta_i}{dt}} = r_i \frac{d\theta_i}{dt}$$

$$v_i^{mt} = r_i \frac{d\theta_i}{dt}$$

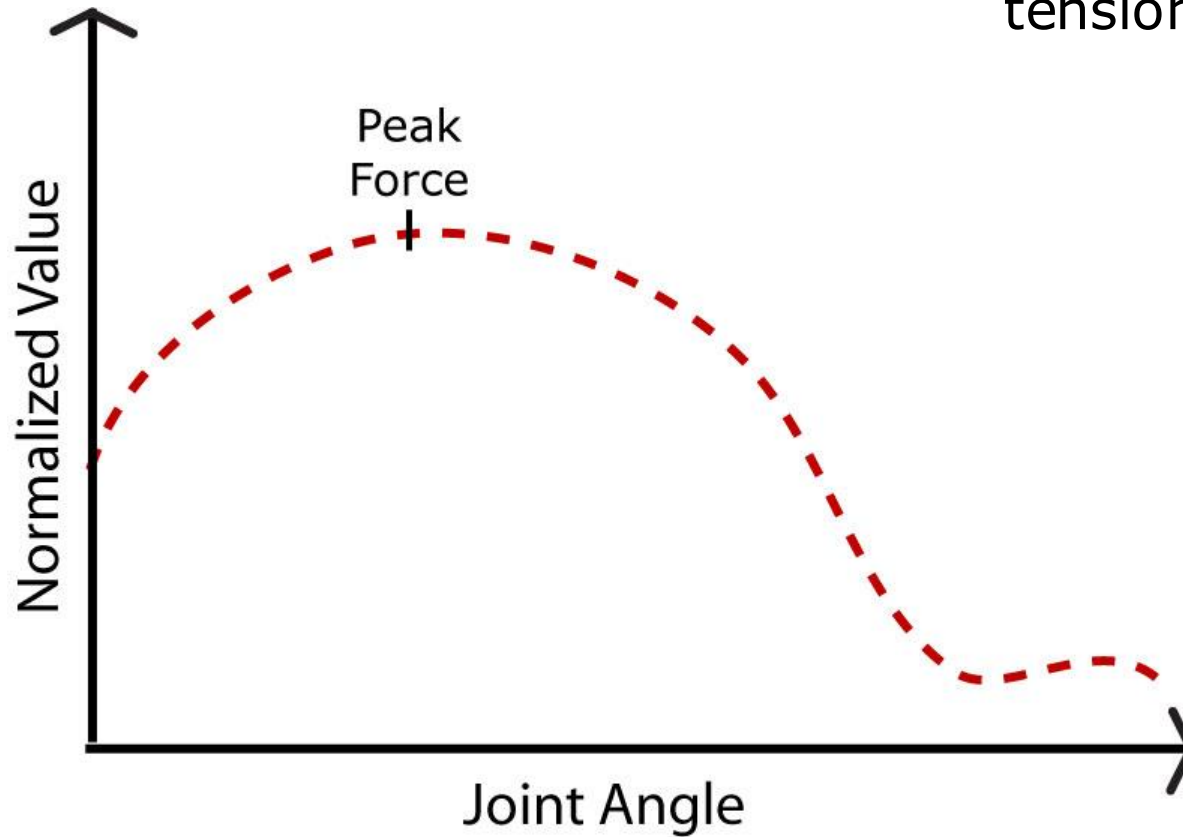
$$v^{mt} = \sum_{i=1}^n r_i \frac{d\theta_i}{dt}$$

Plan for today

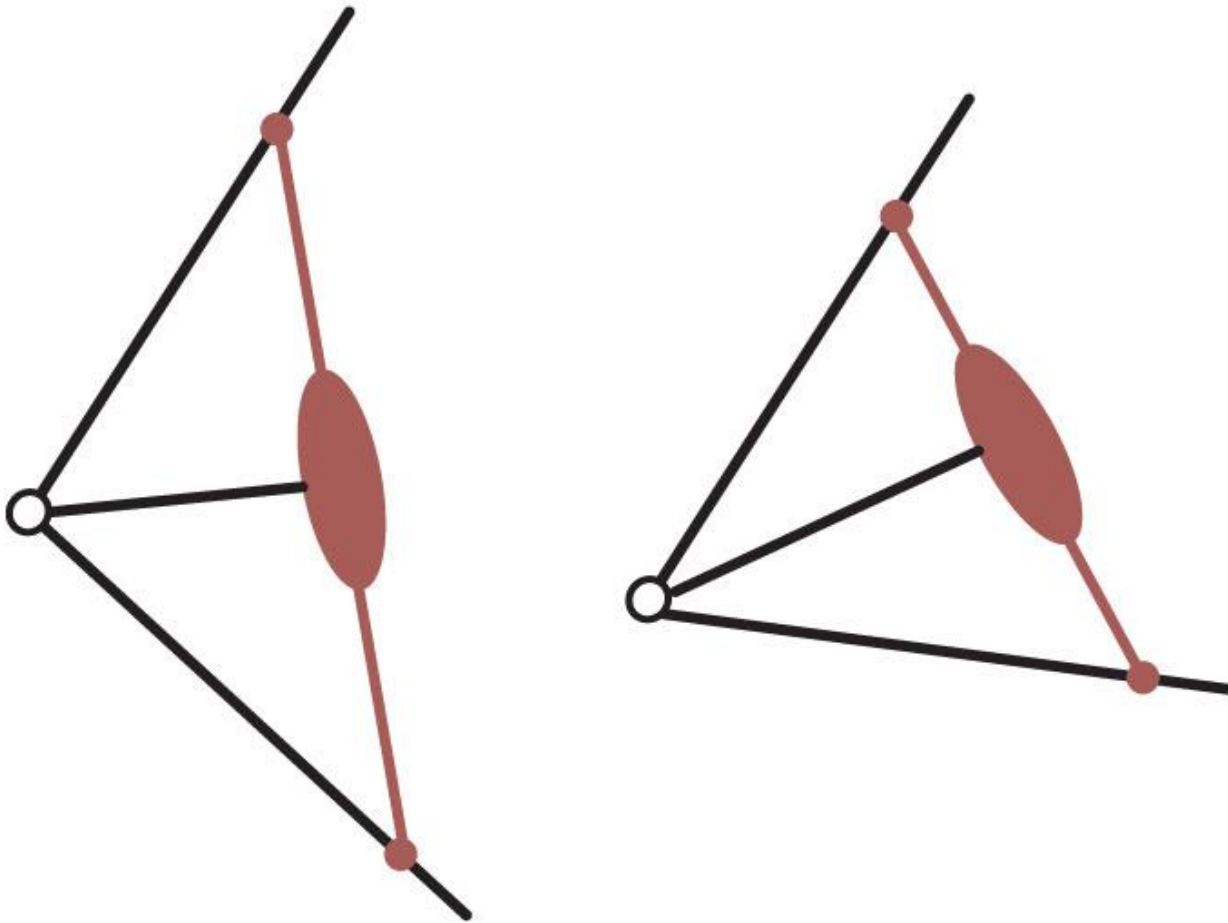
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Forces are not constant

Force follows the length-tension curve



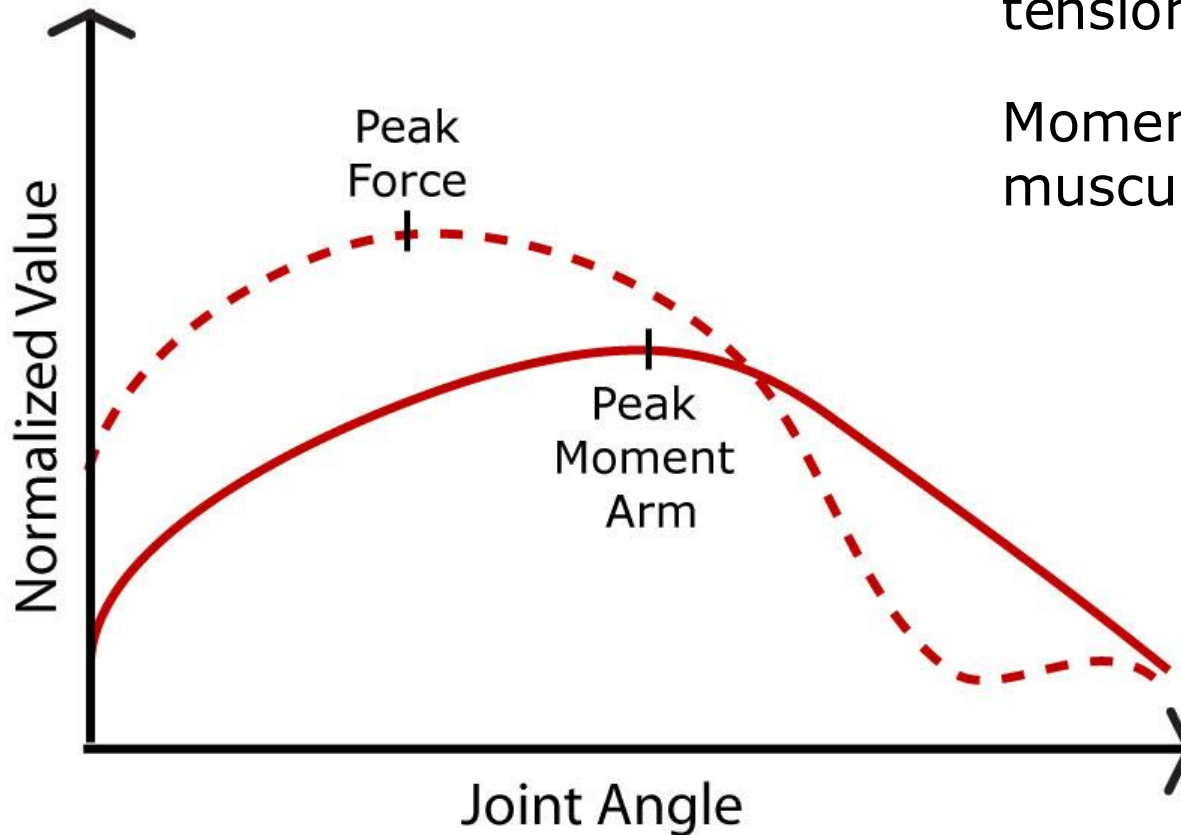
Moment arms are not constant



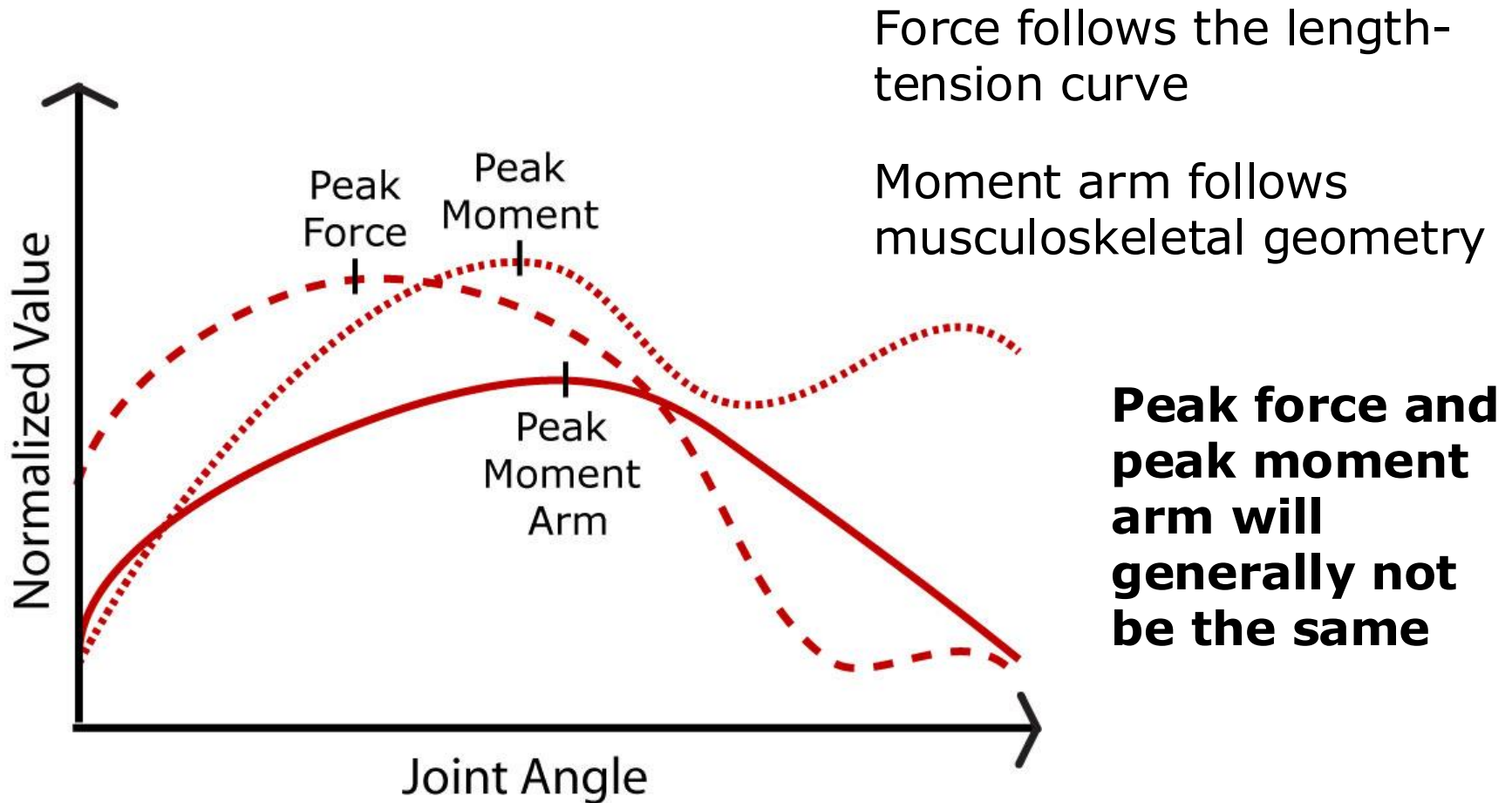
Moment arms are not constant

Force follows the length-tension curve

Moment arm follows musculoskeletal geometry



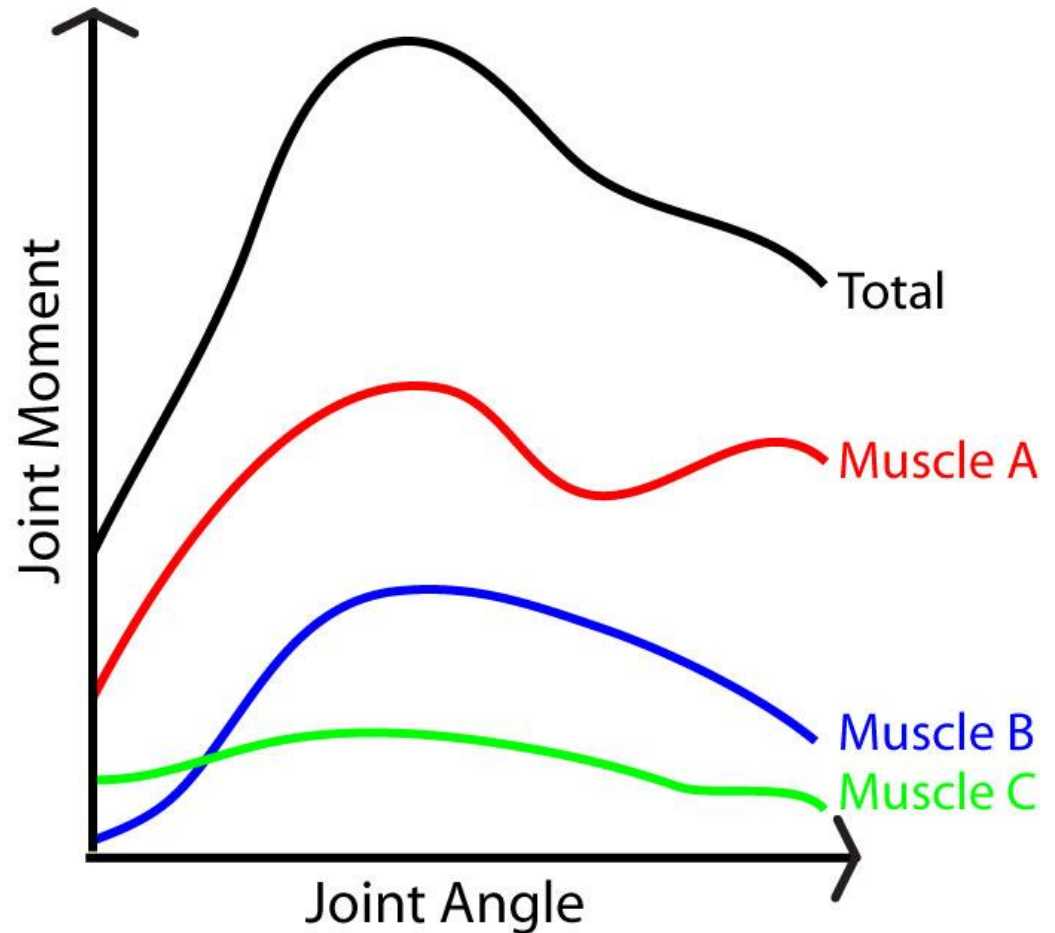
Muscle moment is PRODUCT of force and moment arm



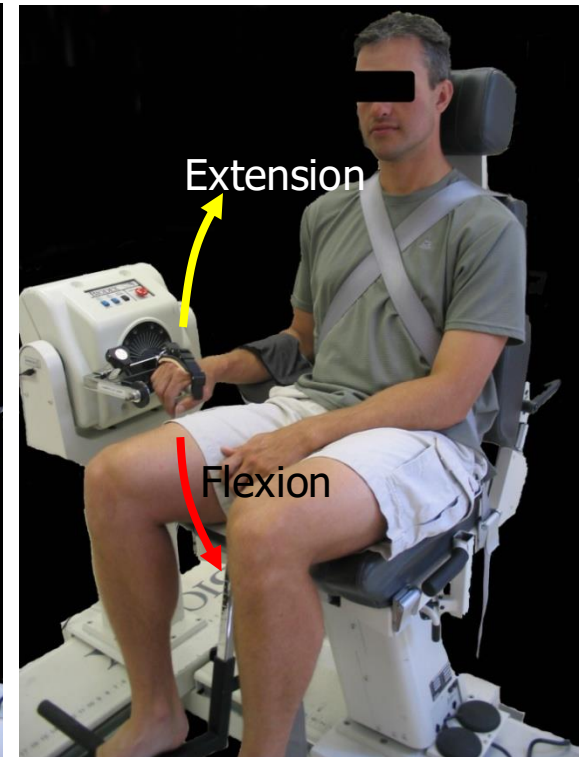
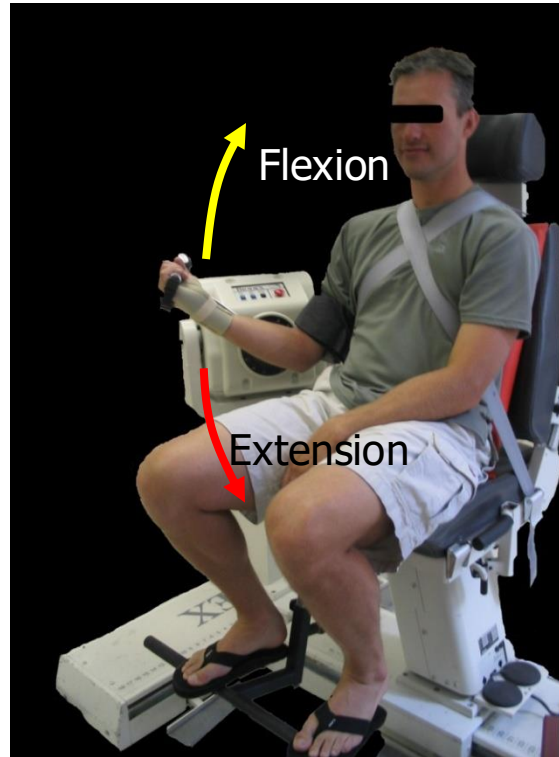
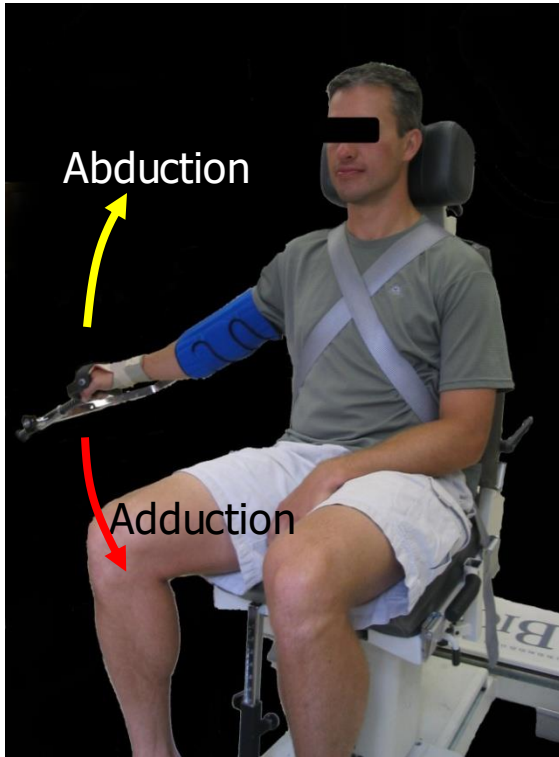
Moments of muscle groups

Normally, more than one muscle is active.

For this case, the moment vs joint angle relationship for all synergists may be complex.



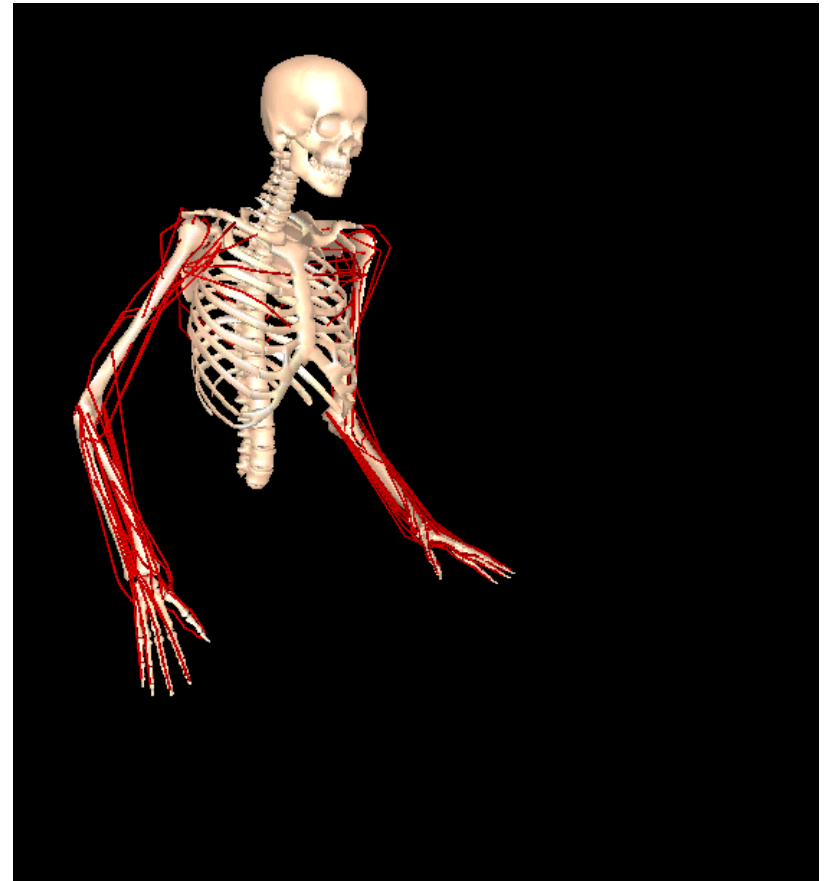
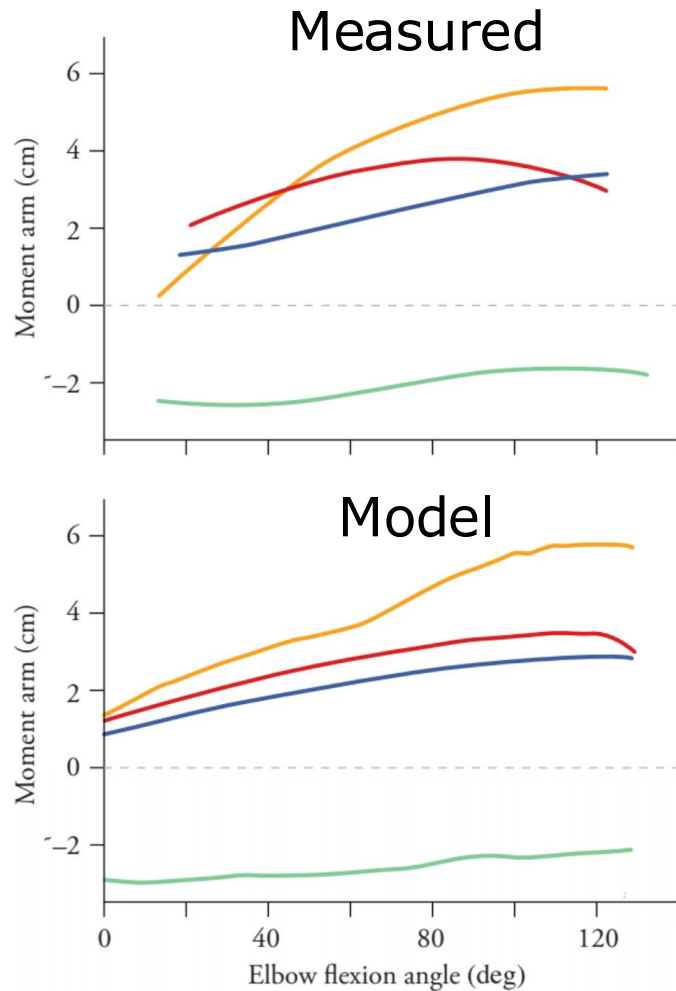
Measuring moments at a joint in humans



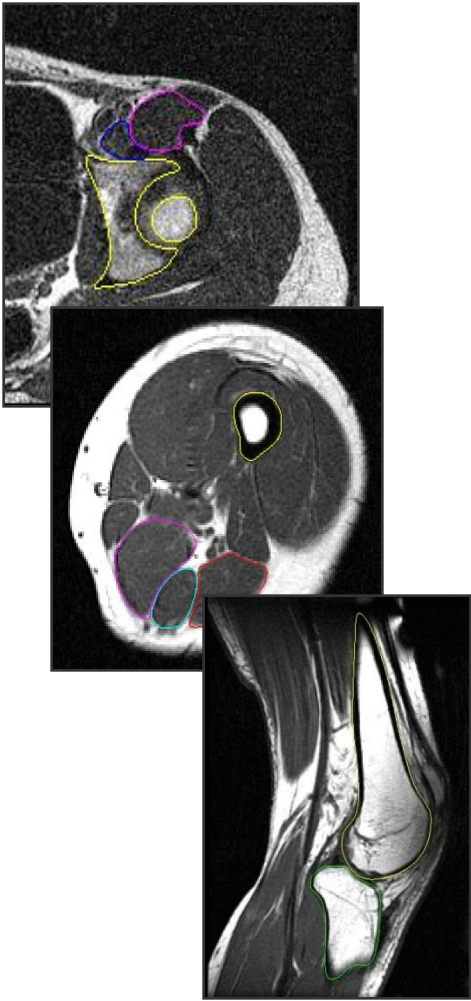
Outline

- ✓ Musculoskeletal geometry
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 - ✓ Example application
- ✓ Moments
 - Modeling of geometry

Measured vs model moment arms



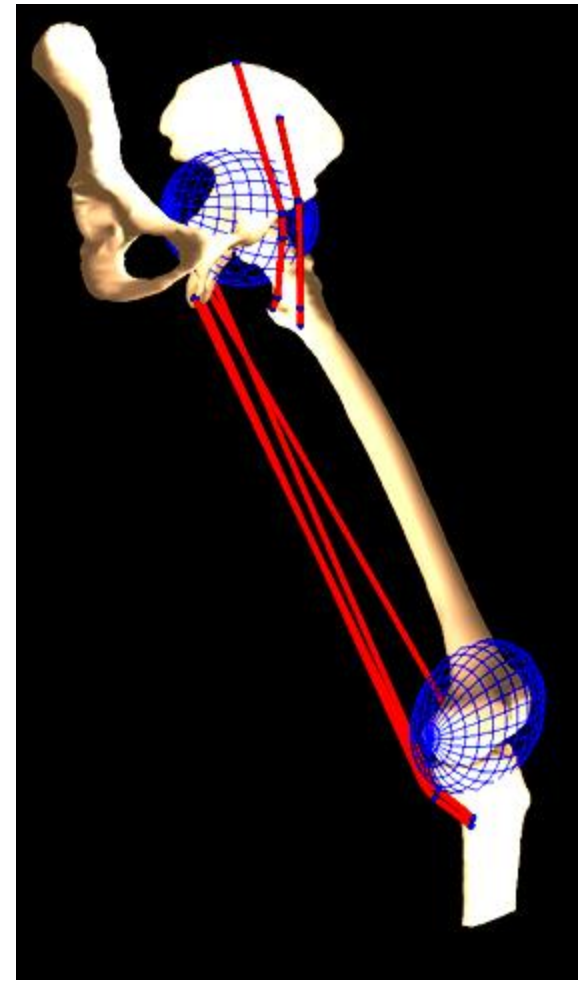
Modeling pipeline



MR images



3D reconstruction



*line muscles combined with
geometric assumptions*

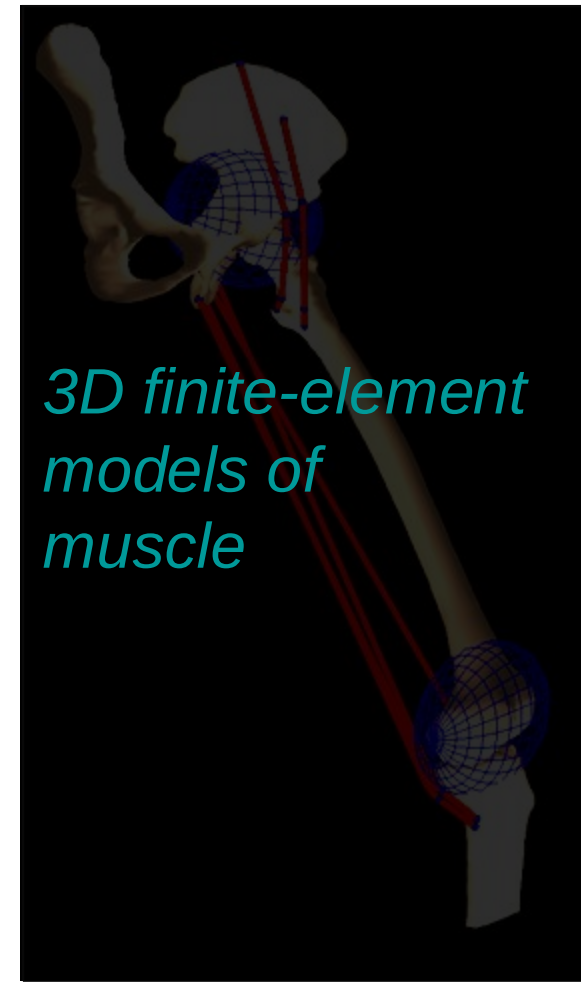
Future modeling pipeline



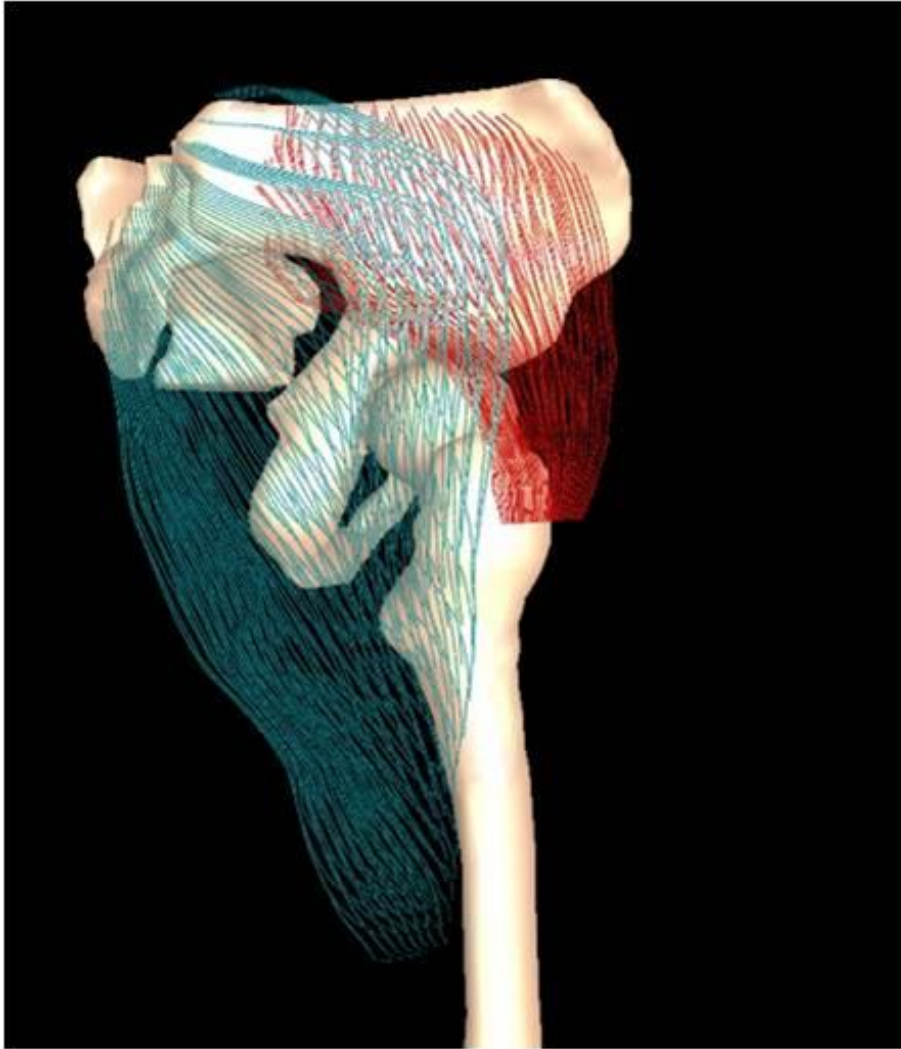
MR images



3D reconstruction

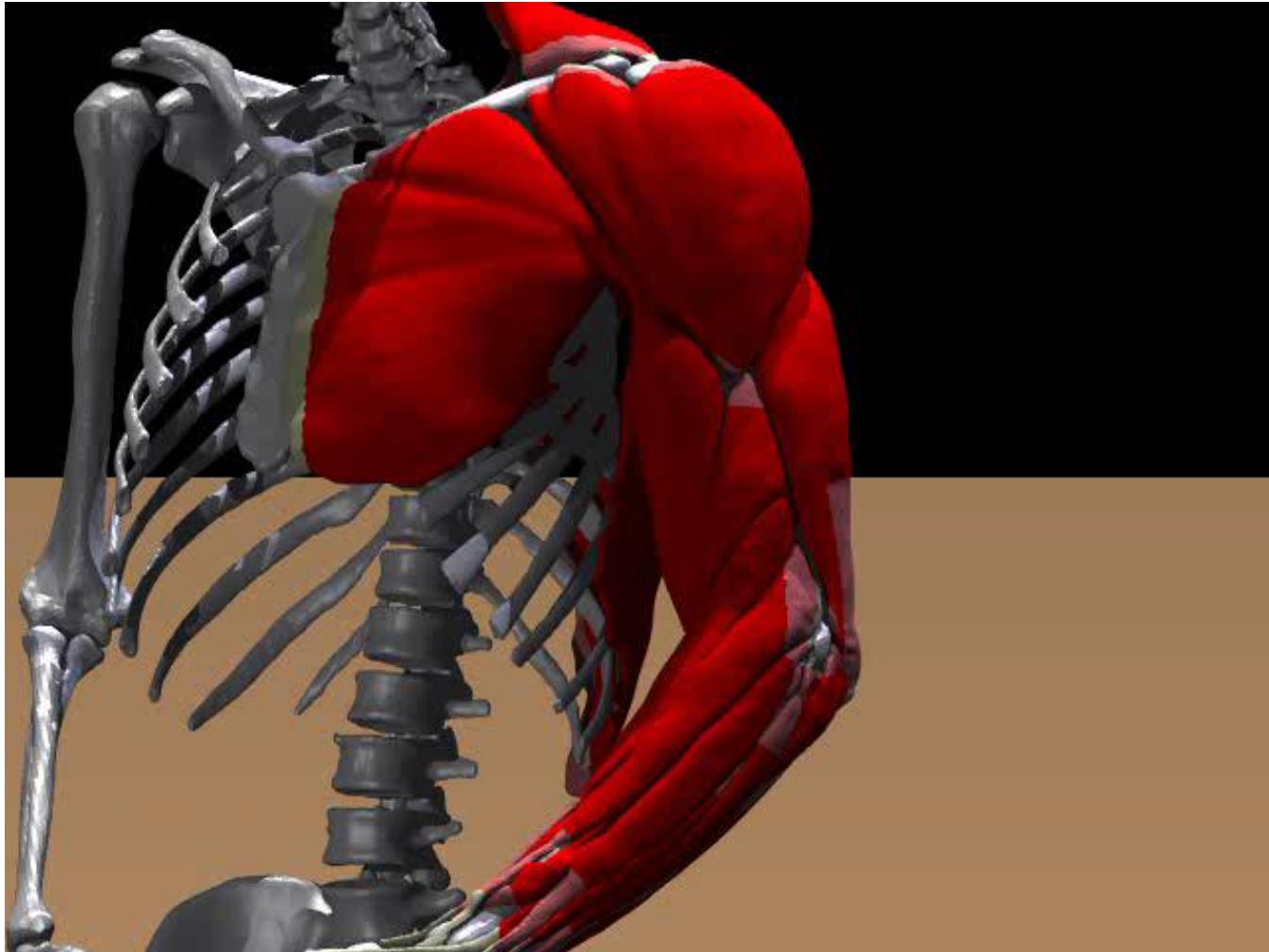


Gluteus maximus and medius



Blemker et al., 2005

Many muscles, high speed



Summary

- Questions we can answer:
 - What are moments and moment arms?
 - How do we measure moments and moment arms?
 - How do we model musculoskeletal geometry?
- Problems we can solve:
 - Calculating muscle force from measurements of moment and moment arm
 - Understanding how moment arm changes muscle function